Network Canvas Architect: From concepts to practices

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Why Networks?

- Link micro-level behaviours to emergent macro-level outcomes (“Coleman’s boat”).
- Consider individuals in their social context.
- Understand emergent phenomena, such as swarming, imitation, diffusion.
- Represent social life with high fidelity, empirically-grounded relationships.

Canonical Example 1

The strength of weak ties

- People tend to want their close friends to know each other.
- As they ‘close these triads’ they create clustering.
- Information circulates faster within clusters.
- So to find novel information, such as a job opening, we can look to the unclosed triads of weak ties rather than the highly clustered strong ties.

Canonical Example 2

The six degrees of separation

- Everyone in the world is connected to at least someone else by virtual of birth.
- On average, how many hops would it take to get from one to another.
- Milgram asked people to send a package from Nebraska to Boston.
- The median chain of referrals was six. Hence the six degrees of separation.

Networks as a Social Construction

- Networks constitute a method and a theory
- The world is not made of networks, but regularities that can often be described as such.
- Networks have persisted
  - Because of their analytic power
  - Because of their synthetic utility

Basic Terms

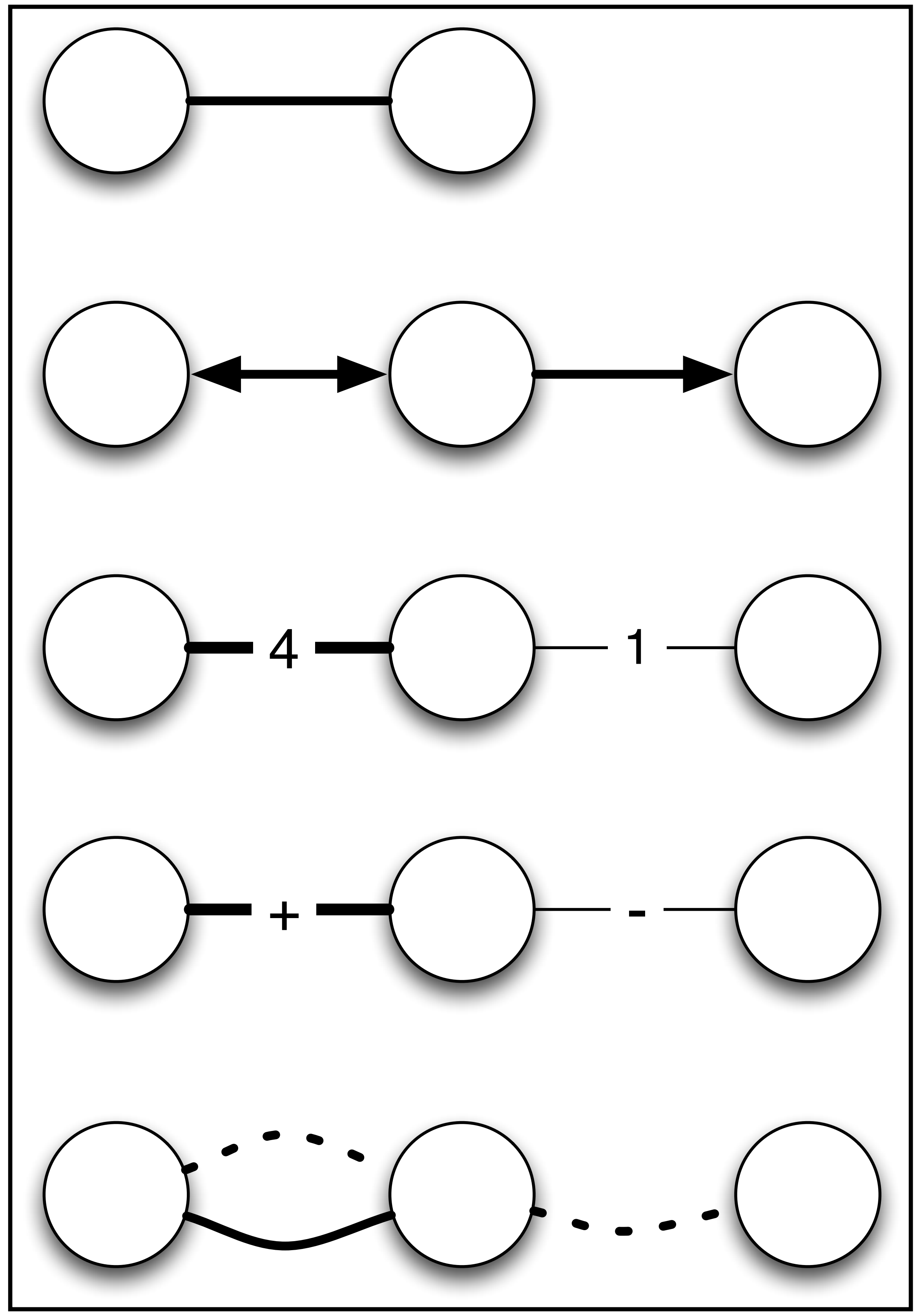
- ***Node (or vertex)***: Basic unit of analysis. Usually of similar type. Can be any boundable object: people, countries, organizations, departments, email messages.

- ***Tie (or edge / arc)***: relation connecting two nodes.

- ***Mode***: The type of relations. One-mode mean all nodes type. Two-mode means two kinds of nodes (such as authors and journals)

- In mathematical notation, G is a graph. G is also a set that contains V vertices and E edges. It can be expressed as
    - G = (V,E)

What’s in a relation?



Undirected: “Edge” - e.g., Facebook friends. $V_{ij} = V_{ji}$

Directed: “Arc” - e.g., Gives advice. $V_{ij} \neq V_{ji}$

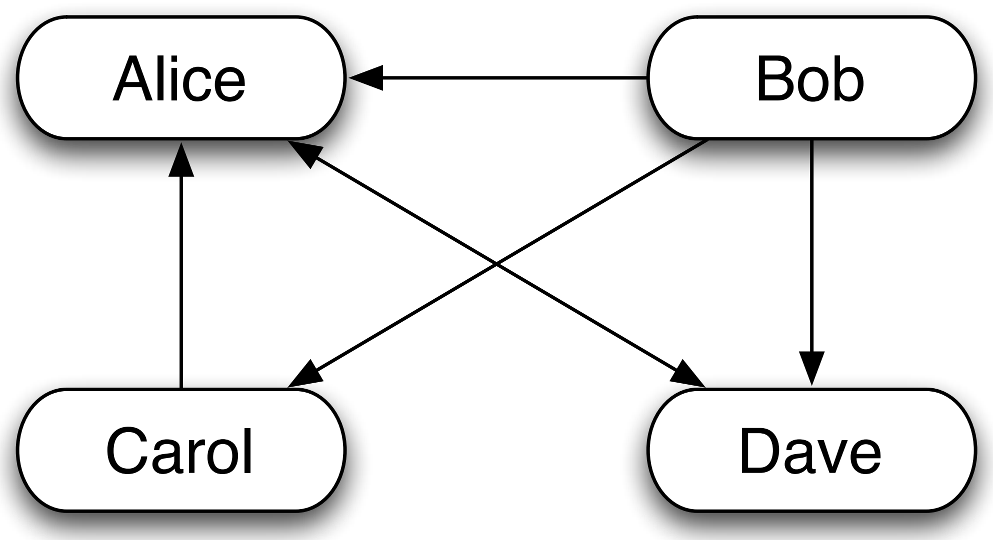
Weighted: Numerical value of relation, e.g., number of messages sent. Higher value = “stronger” tie between people.

Valued: Positive or negative, e.g., Emotional energy; positive / negative reviews on eBay.

Multiplex: Multiple relation types between people.

Representing Network Data

	Alice	Bob	Carol	Dave
Alice	-	0	0	1
Bob	1	-	1	1
Carol	1	0	-	0
Dave	1	0	0	-

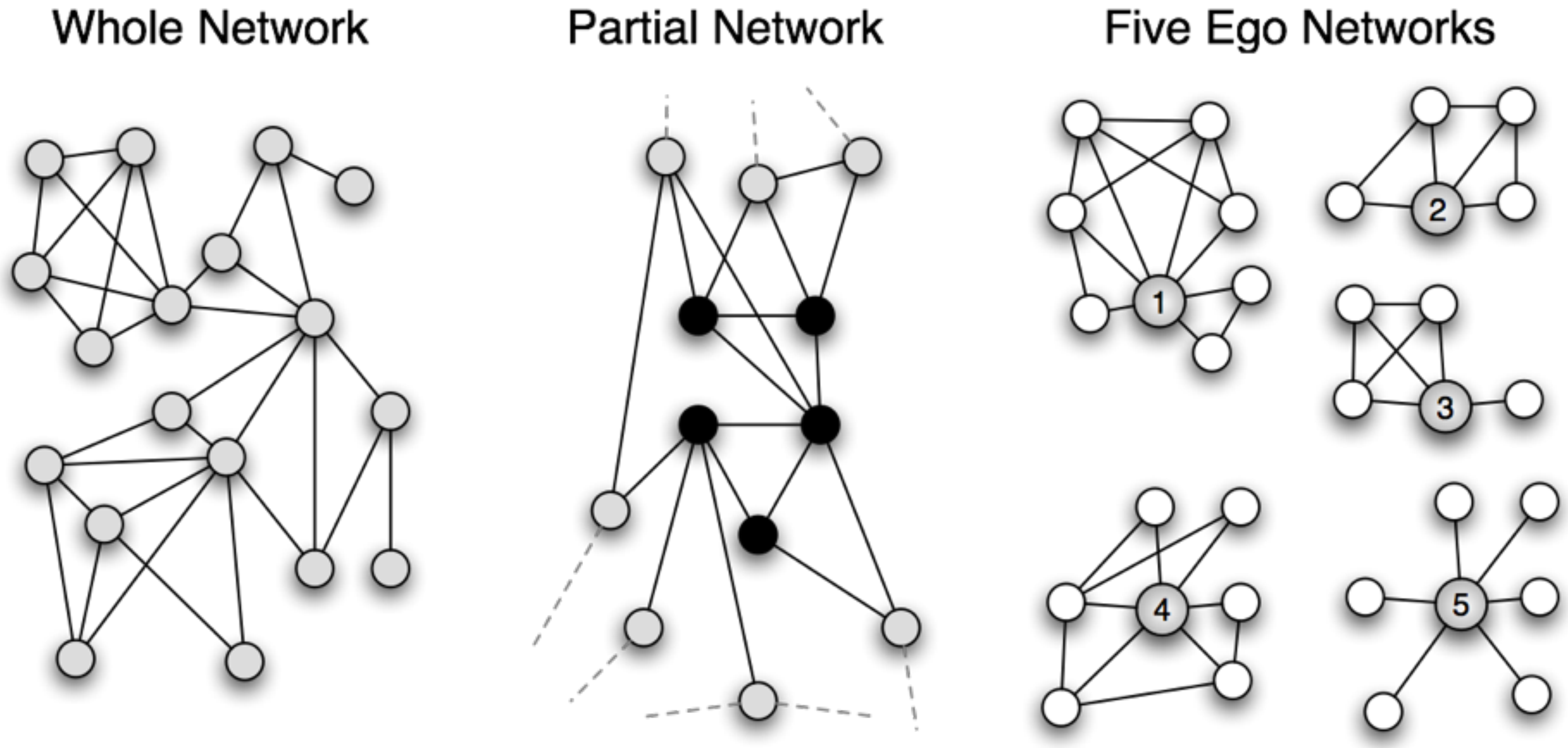


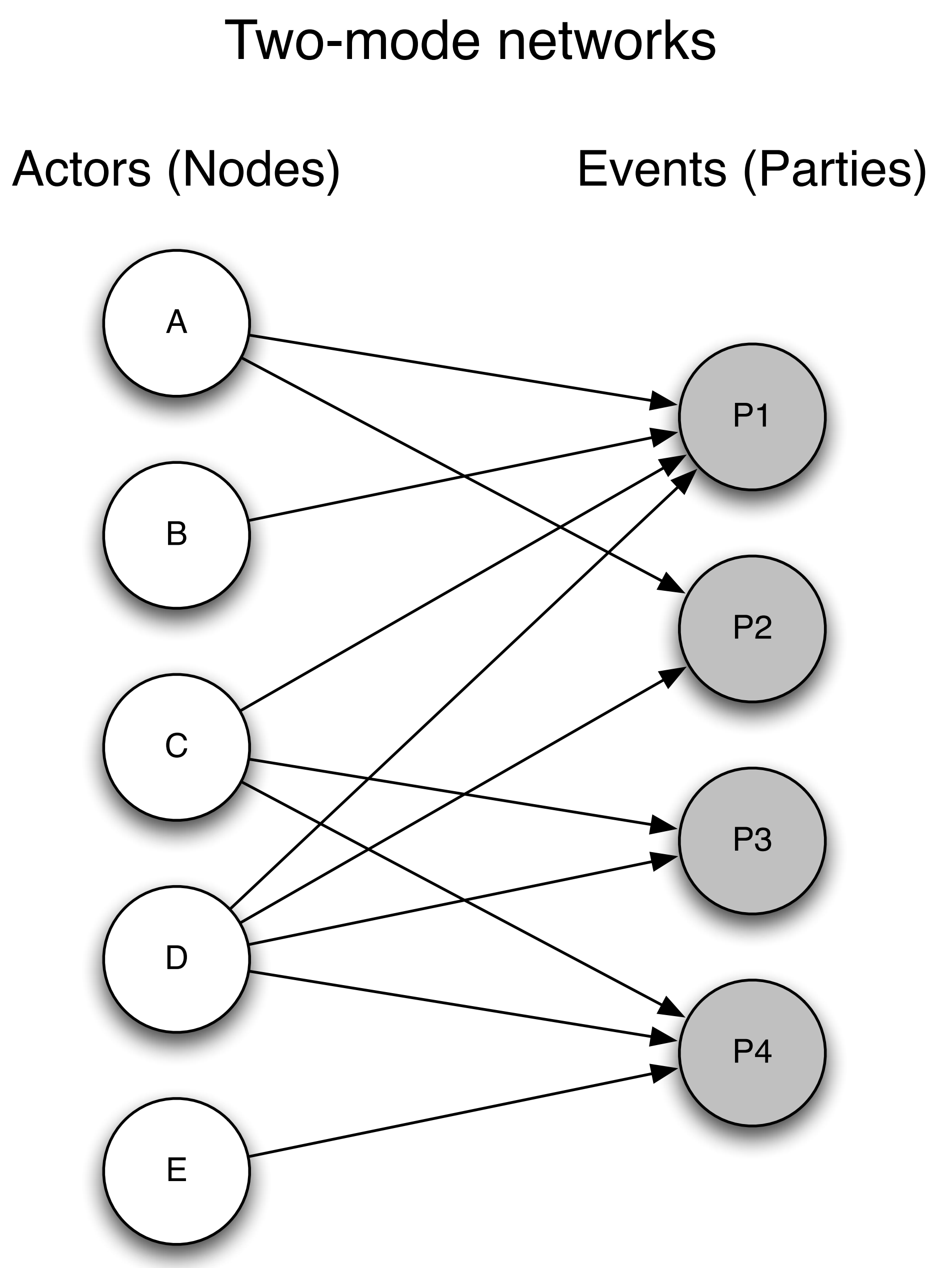
From	To
Alice	Dave
Bob	Alice
Bob	Carol
Bob	Dave
Carol	Alice
Dave	Alice
Dave	Bob
Dave	Carol

Network links can be represented as:

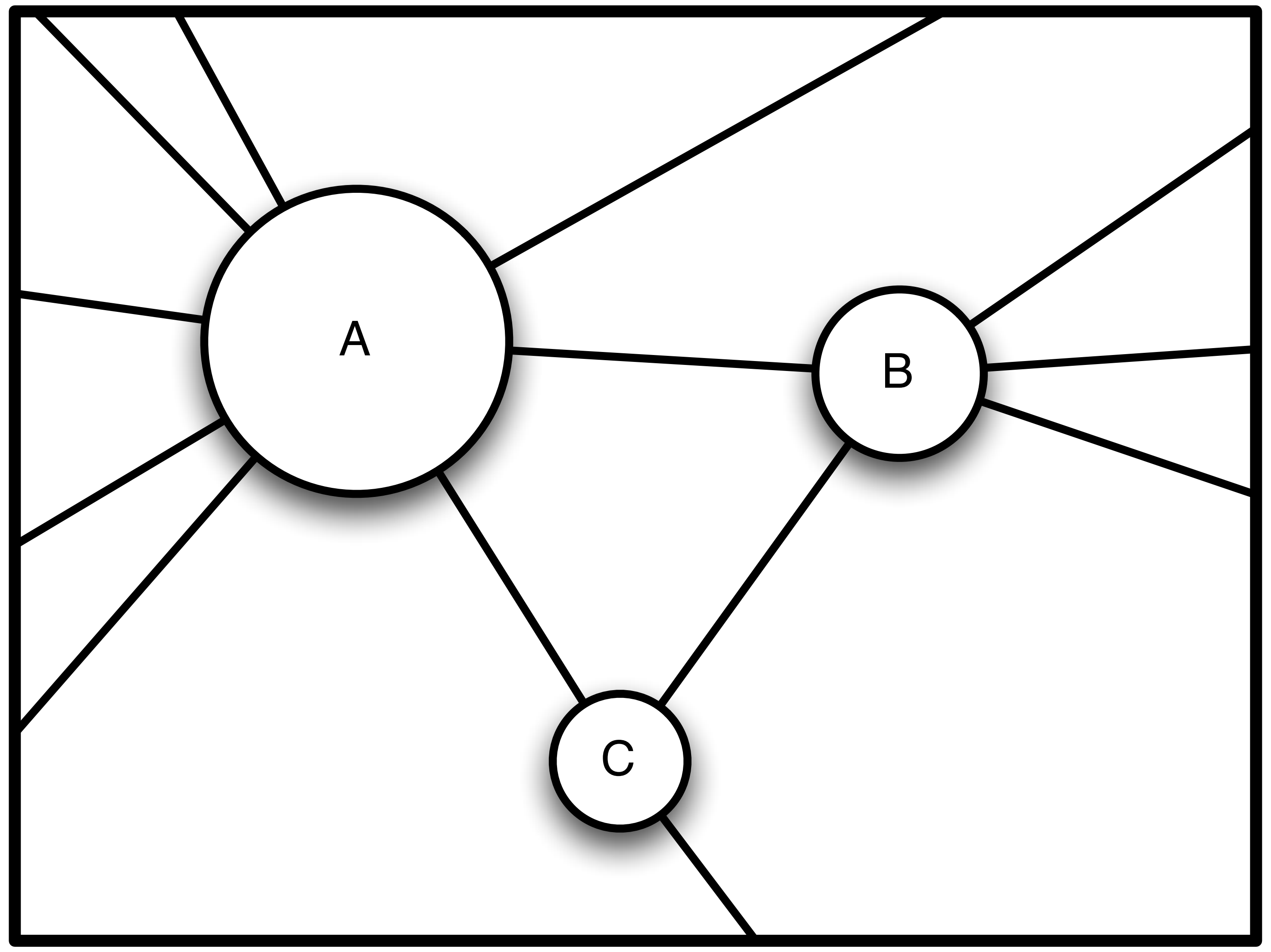
- Adjacency matrices: *from* in rows and *to* in columns,
- Sociograms: relations represented visually,
- Edge lists: *from* in column 1 and *to* in column 2, plus attributes in additional columns.

Network collection types





Units of analysis: Position

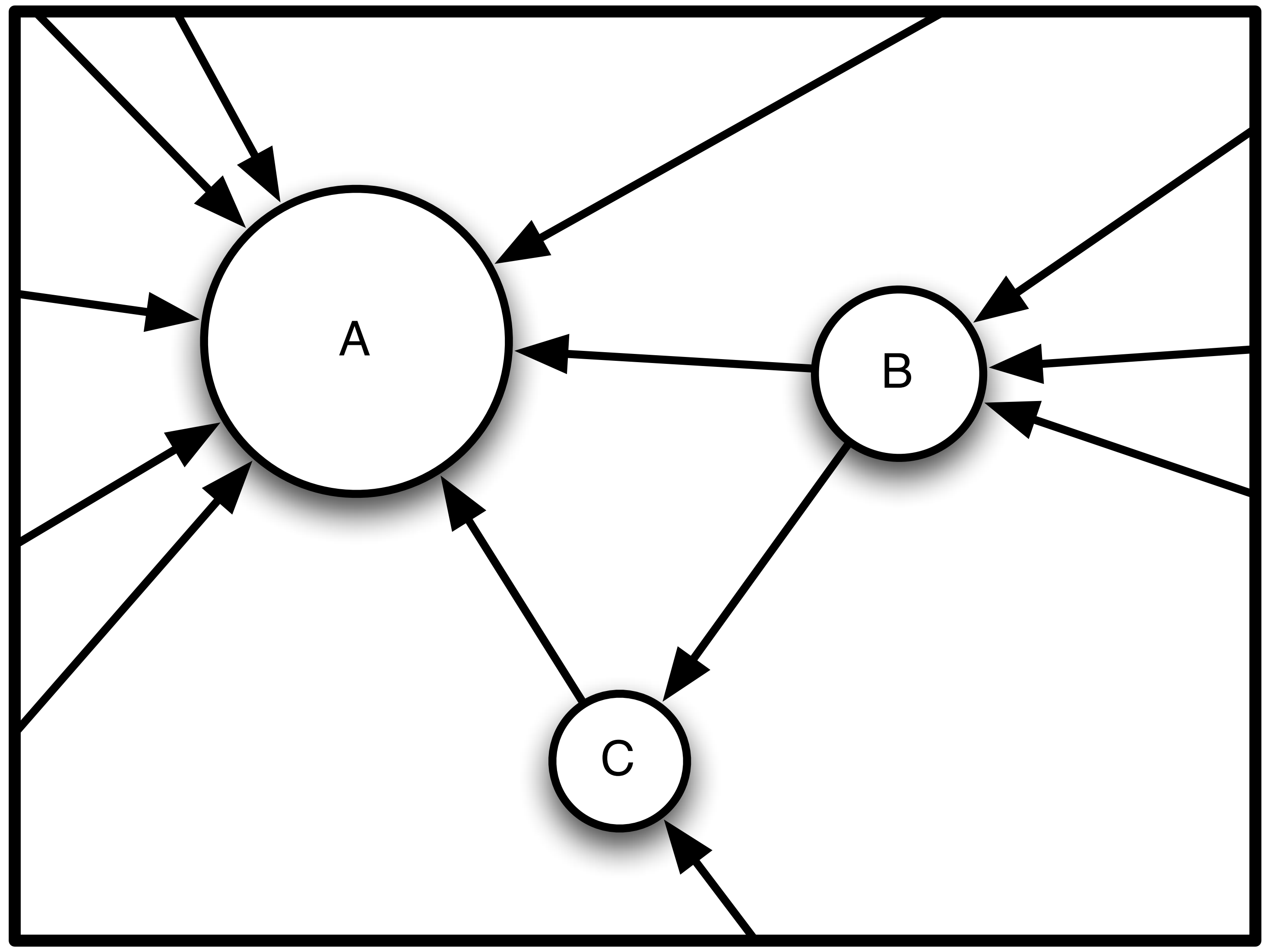


Position:
Looking for a node's position is one of the firsst and most sensible tasks for the analyst.

Nodes with more links are considered more *central.*

Degree centrality is a score for more links, where a link is considered a 'degree'.

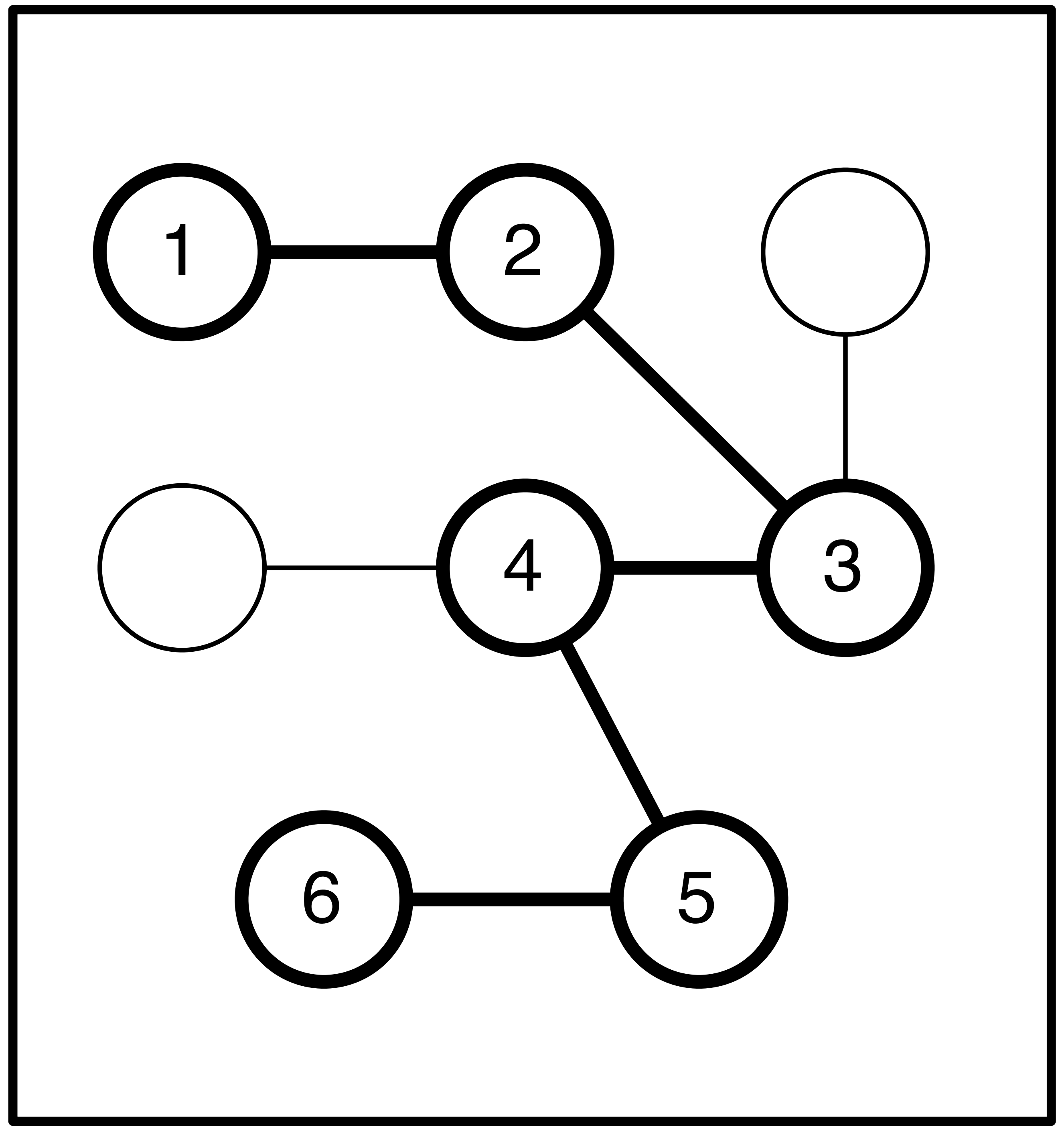
Units of analysis: Position



In- and out-degree:
Some links are directed.

When links represent a message, nodes with a higher *in-degree* are more popular. Nodes with more *out-degree* are more authoritative.

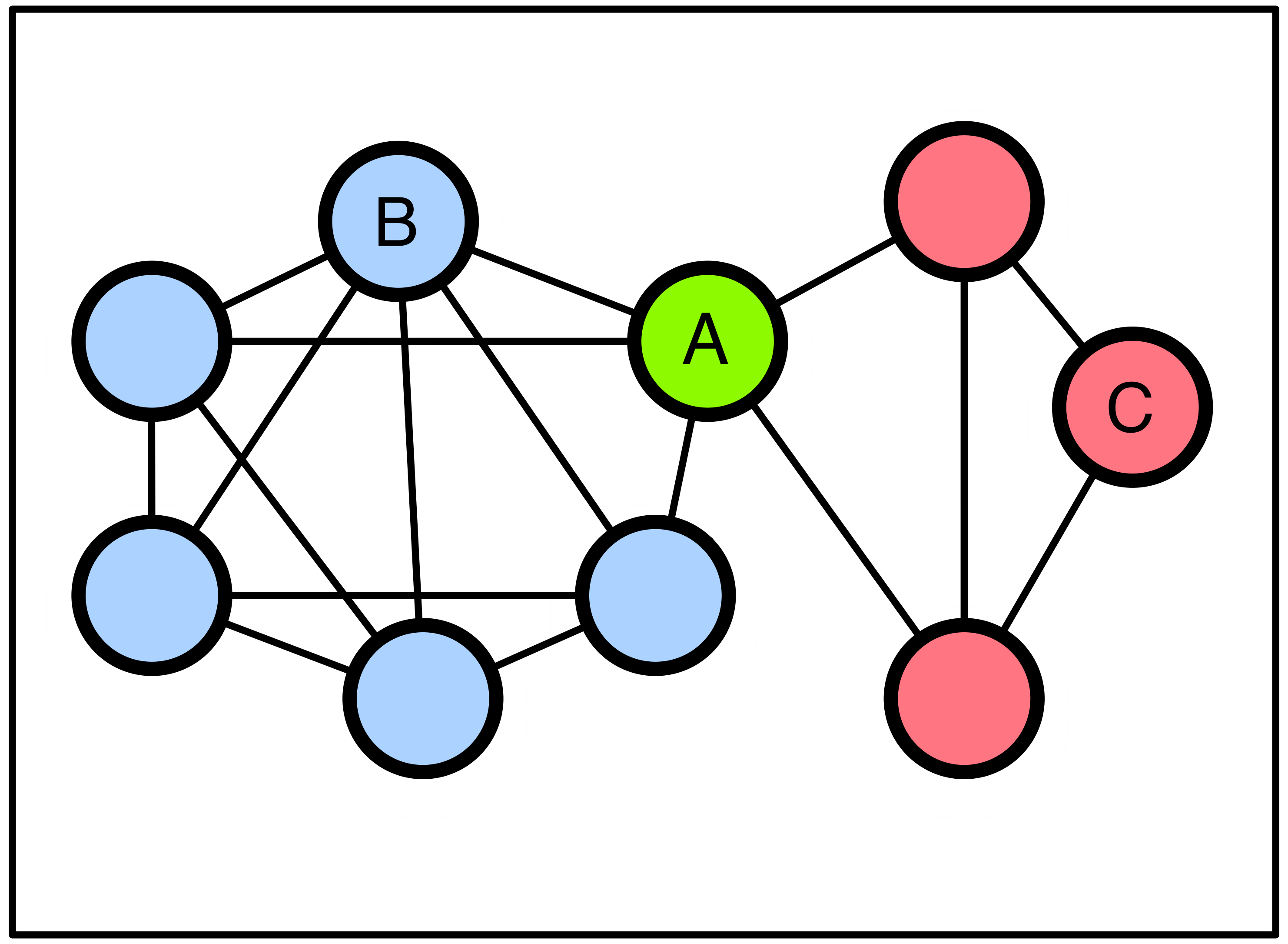
Definition: Geodesic



A geodesic:
Is the shortest path between two nodes.

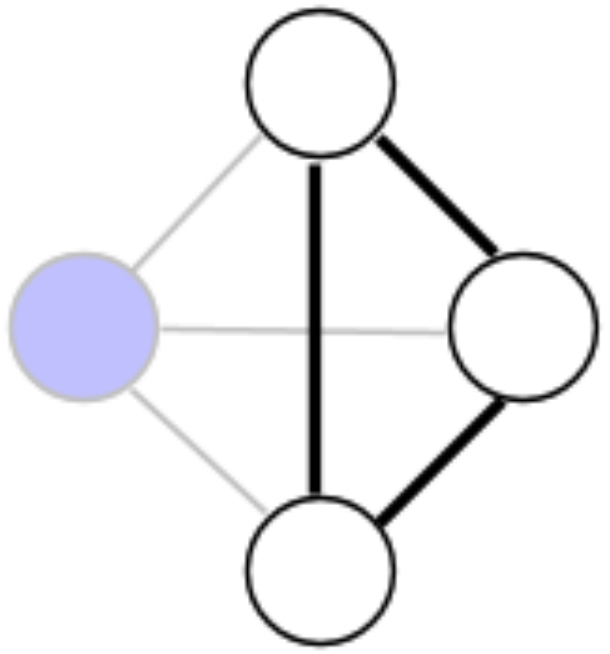
A diameter:
Is the longest of these short paths in the network.

Units of analysis: Position

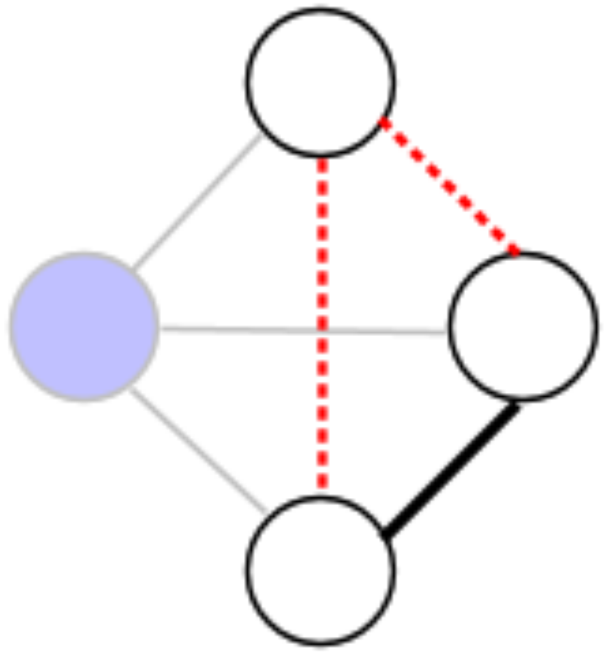


Betweenness:
Being more popular is not always the best measure of centrality. What if you are the only link between two sources of information?

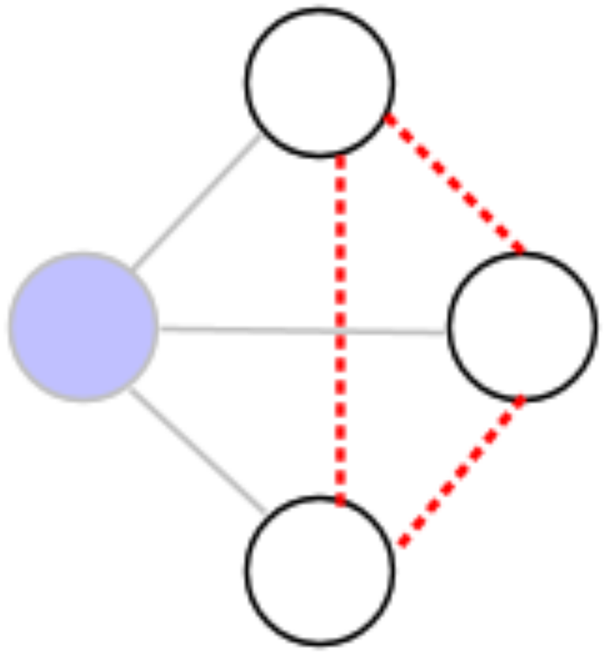
Betweenness centrality measures where individuals lie on shortest paths.



c = 1



c = 1/3



c = 0

Clustering Coefficient

Percentage of all triads that are realised as triples.
View the diagram from the point of view of the purple node.

Local clustering coefficient:
Percent of all nodes around a given node. Acts as a measure of local density.

Global clustering coefficient:
Average of all LCC scores across the entire network.

Transitivity:
(Number of triads *3) / number of triples. This is like the LCC except it's calculated for all the network rather than for all the egos and averaged.

How to make a network?

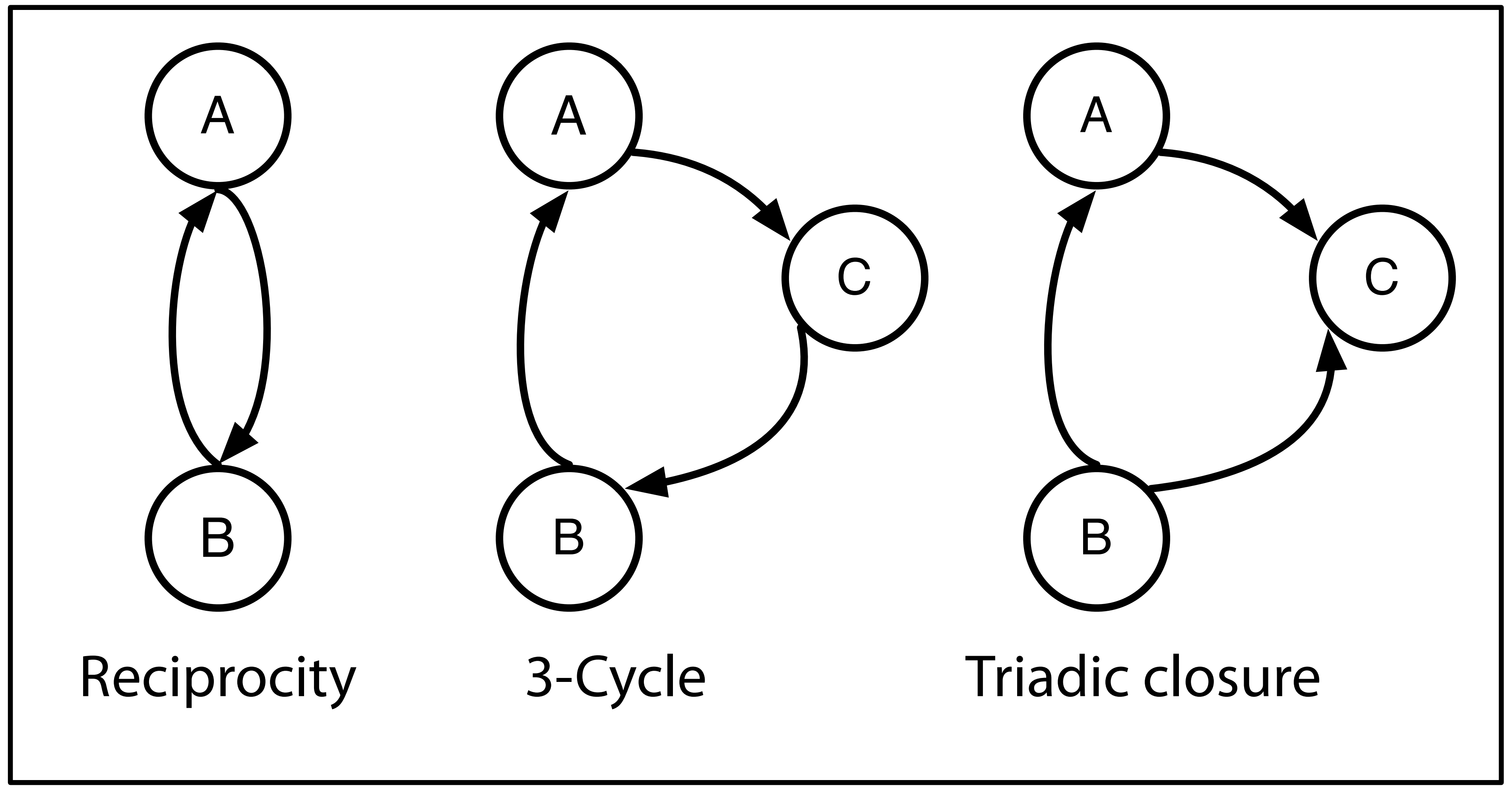
- While some networks can be derived from empirical data or manifested in socio-technical systems, others are synthetic graph objects. Some examples:

  - A star graph is a graph with one centre node and *n* nodes that connect to the centre. (`Star` in igraph)

  - A Bernoulli graph / Erdos-Renyi graph is a graph where edges exist with probability *p* given all possible edges. (`Erdos_Renyi` in igraph)

  - A Molloy graph is a randomly generated graph where the number of nodes are fixed as is the *degree distribution*. (`rewire` in igraph)

Units of analysis: Micro

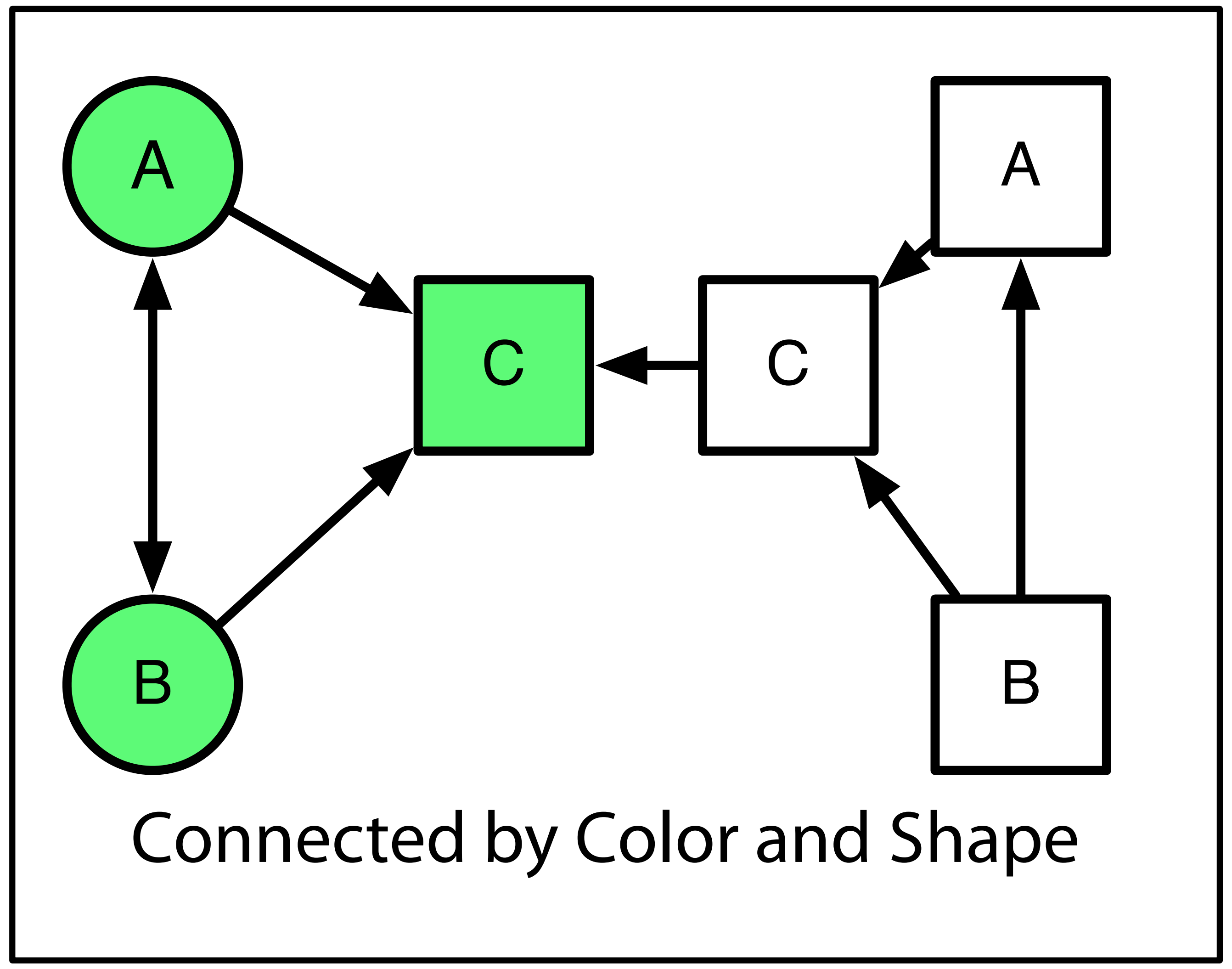


Configurations:
Certain arrangements of nodes are important, such as reciprocal edges, 3-cycles and tradic closure.

Reciprocity is extremely prevalent in networks.
3-cycles imply egalitarianism.
Triadic closure is common and implies hierarchy.

Homophily

Units of analysis: Micro

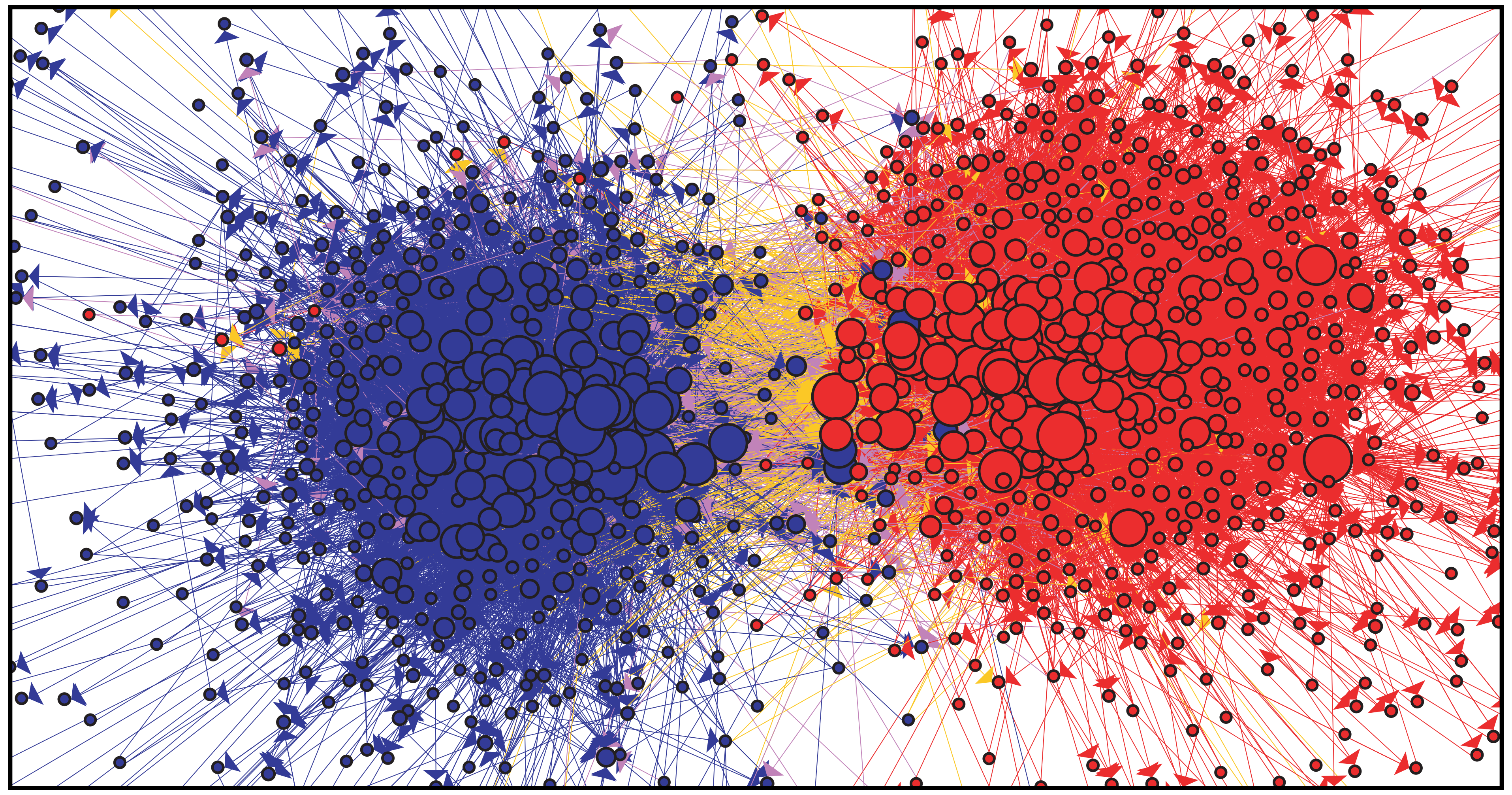


Homophily:
Individuals of like type are particularly prone to linking to one another.

The trick is not finding homophily, but the right kind of homophily.

Baseline homophily is how much you would expect by chance. In-breeding homophily is homophily over and above this value.

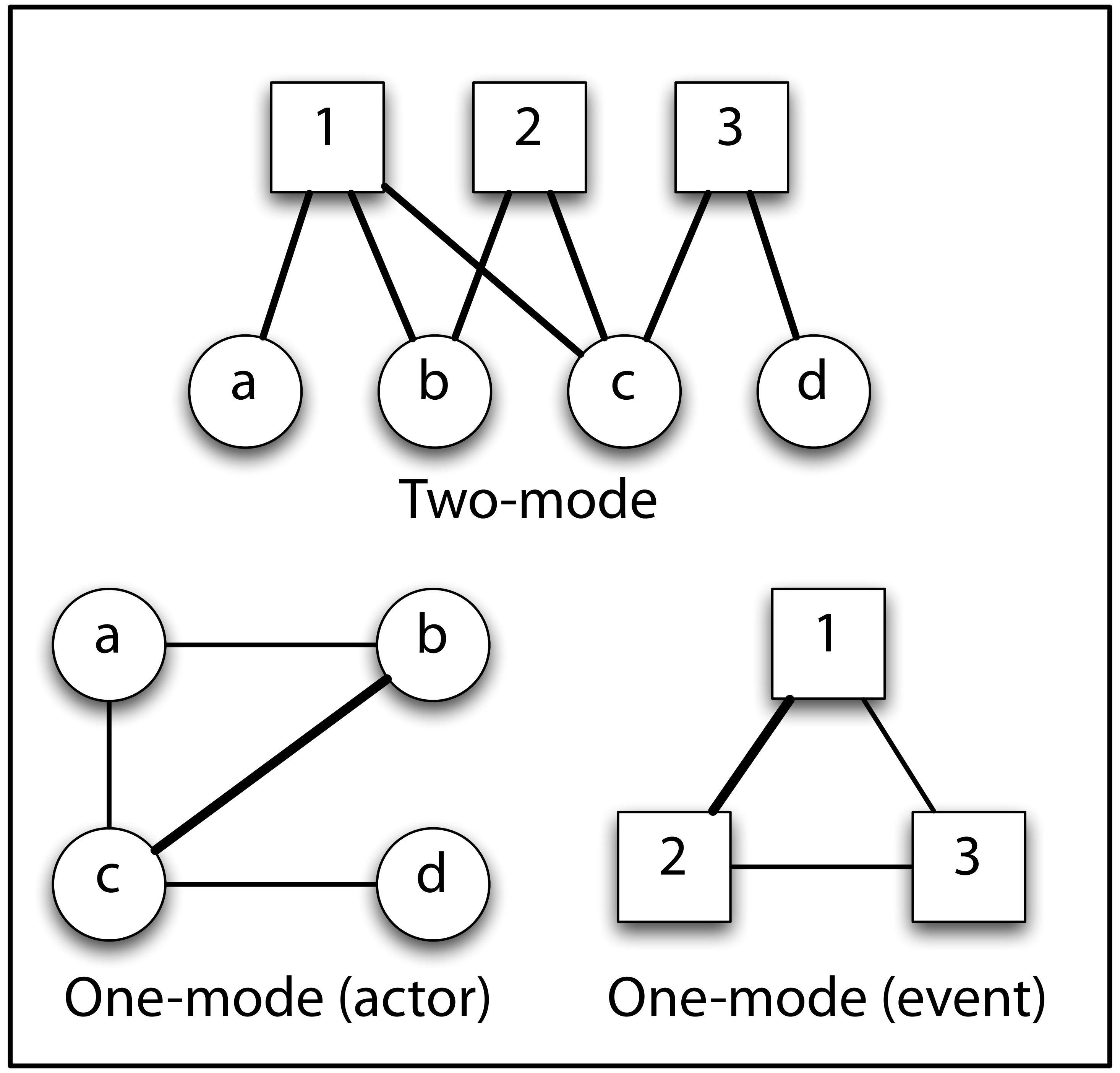
Units of analysis: Meso



Communities:
Clusters of nodes tend to group together. Where there are more nodes within a group than between a group we consider this a community.

Community detection is one of the most common tools in network analysis outside of looking for node position. Communities tend to emerge based on the micro level processes previously discussed.

Networks Data: Modes

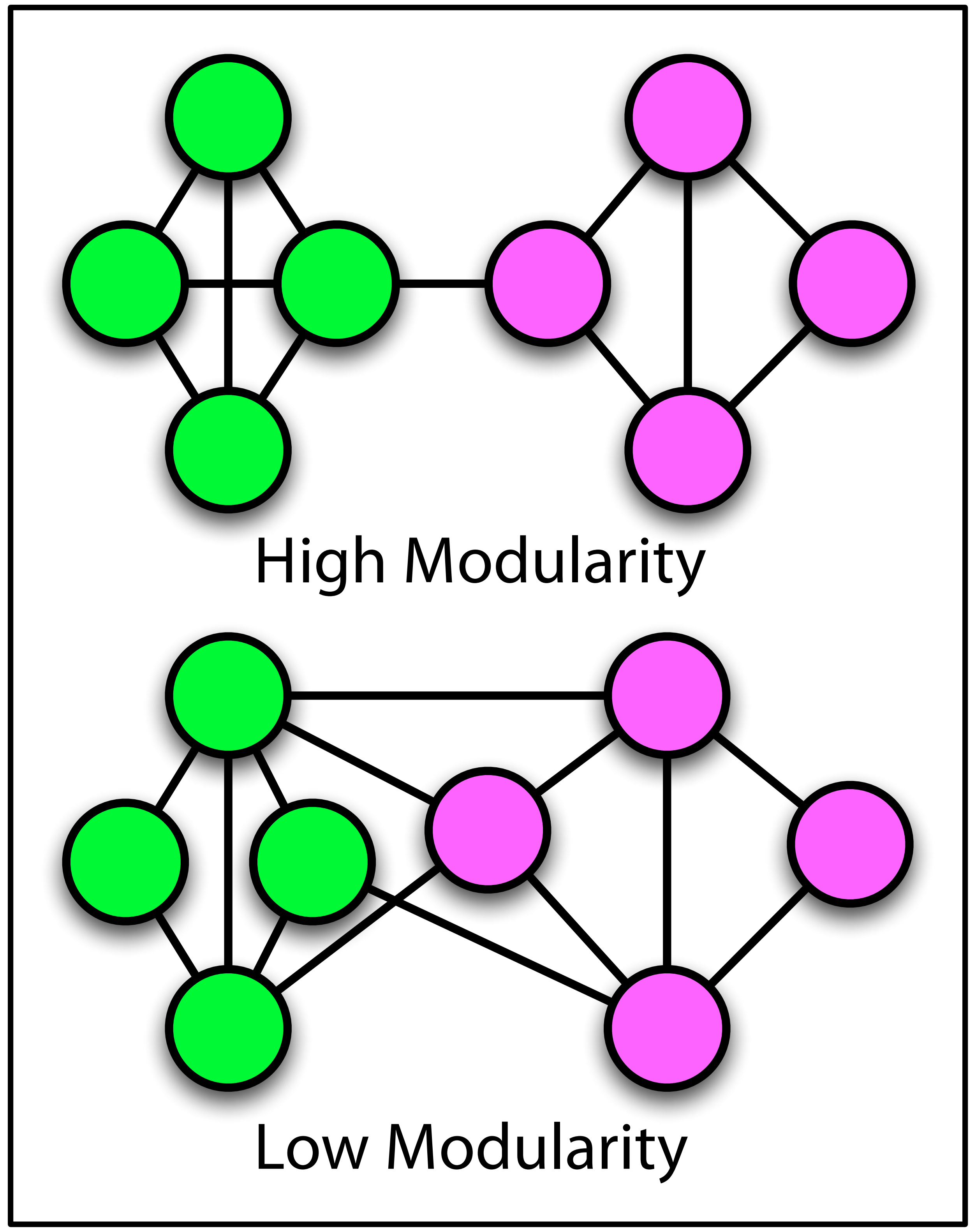


Modes:
Networks of different types. For example, comments on a Facebook page.

How to view the network:
Two modes often make formatting difficult, but show both commenters and posts.
One mode projects show "co-activity".

An edge exists in a co-activity network, if two people comment on the same post, or two posts share the same commenter

Units of analysis: Meso

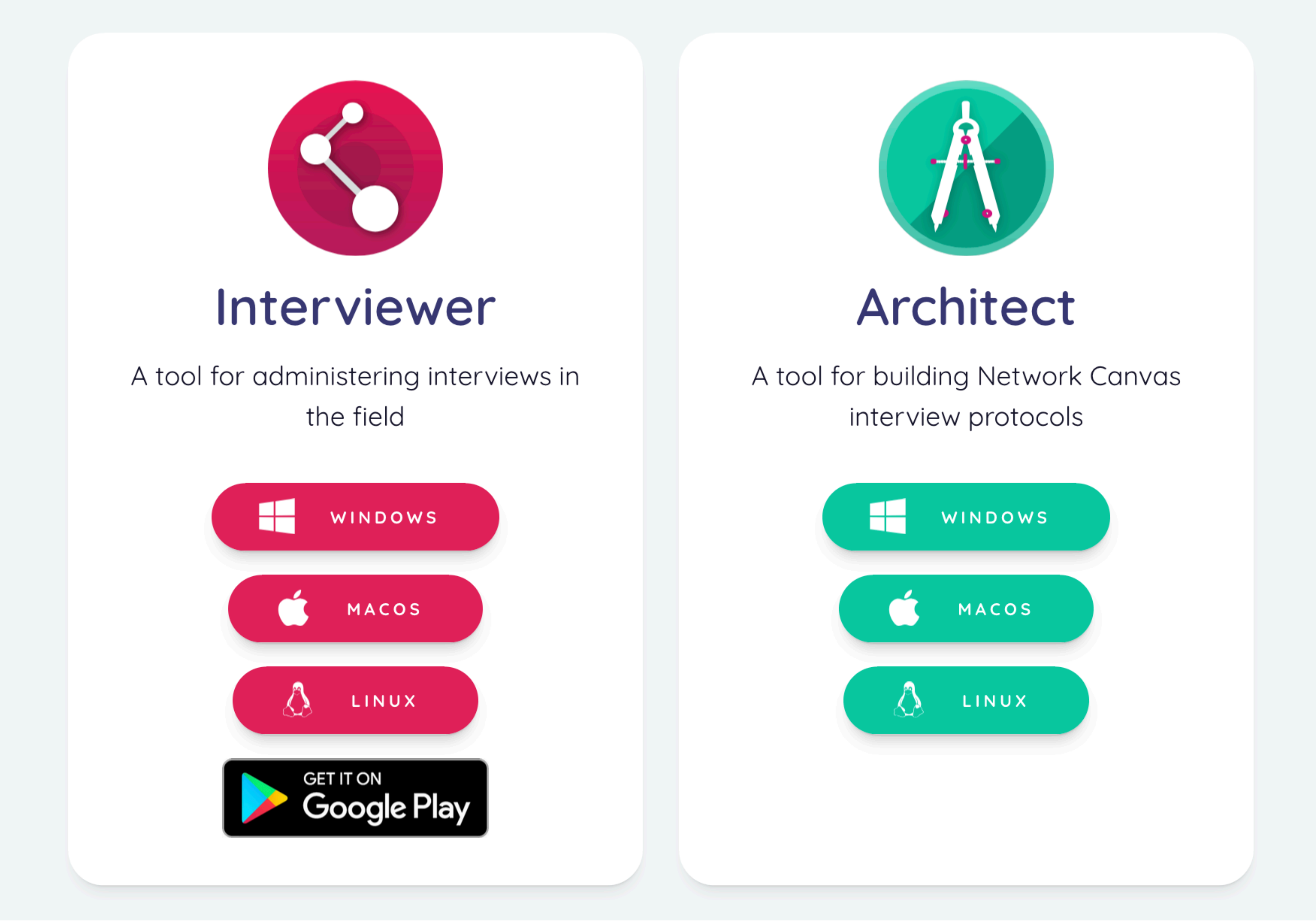


Modularity and Communities:
When you run community detection it is trying to find a solution to maximize modularity (the measure of edges within a group to nodes between groups).

High modularity means clear groups.
Low modularity means fuzzy groups.

A score > 0.3 usually considered a good fit.

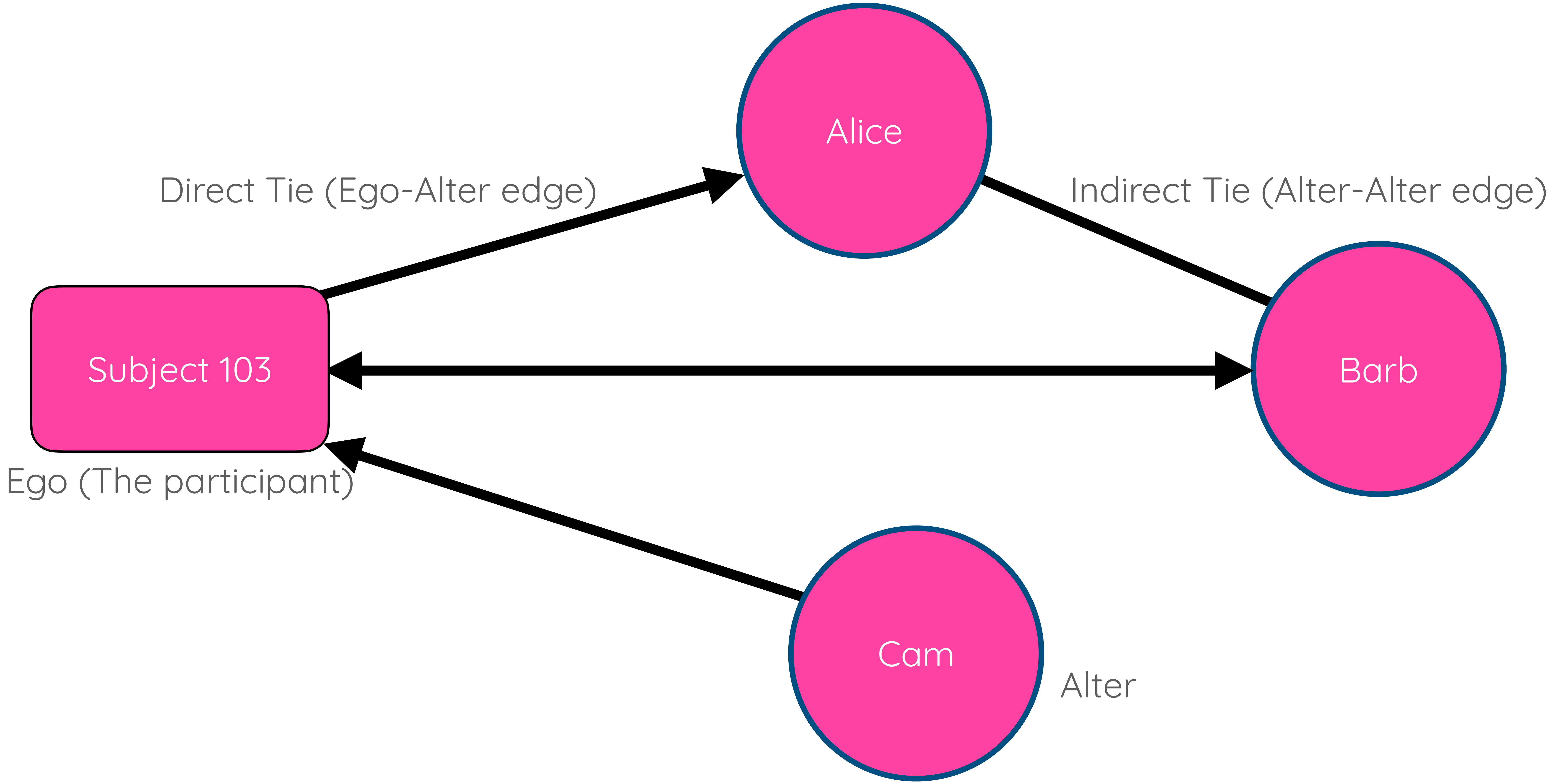
Installing Network Canvas: https://networkcanvas.com/download




Network Canvas
6.4.0
Architect
A tool for building Network Canvas Interviews
Show welcome
Welcome to Architect!
If you are new to the software, please consider watching the overview video to the left. It will explain how the software works, and introduce all the essential skills needed to build an interview protocol. We also have extensive tutorials and information on a range of topics on our documentation website, which you can visit using the link below.
Alternatively, to get started right away use the buttons below to create a new interview protocol, or open an existing one from elsewhere.
WATCH OVERVIEW VIDEO
VISIT DOCUMENTATION WEBSITE
Resume Editing
OSN20_week3b_wholeNetDemo_redacted (schema version 7)
/Users/work/Documents/Teaching/OSN2020/Exercises/OSN20_wee (schema version 7).netcanvas
SB22_architect_demo
/Users/work/Documents/Teaching/Sunbelt/SB22_architect_demo.netcanvas
SB22_workshop_protocol
/Users/work/Documents/Teaching/Sunbelt/SB22_workshop_protocol.netc...

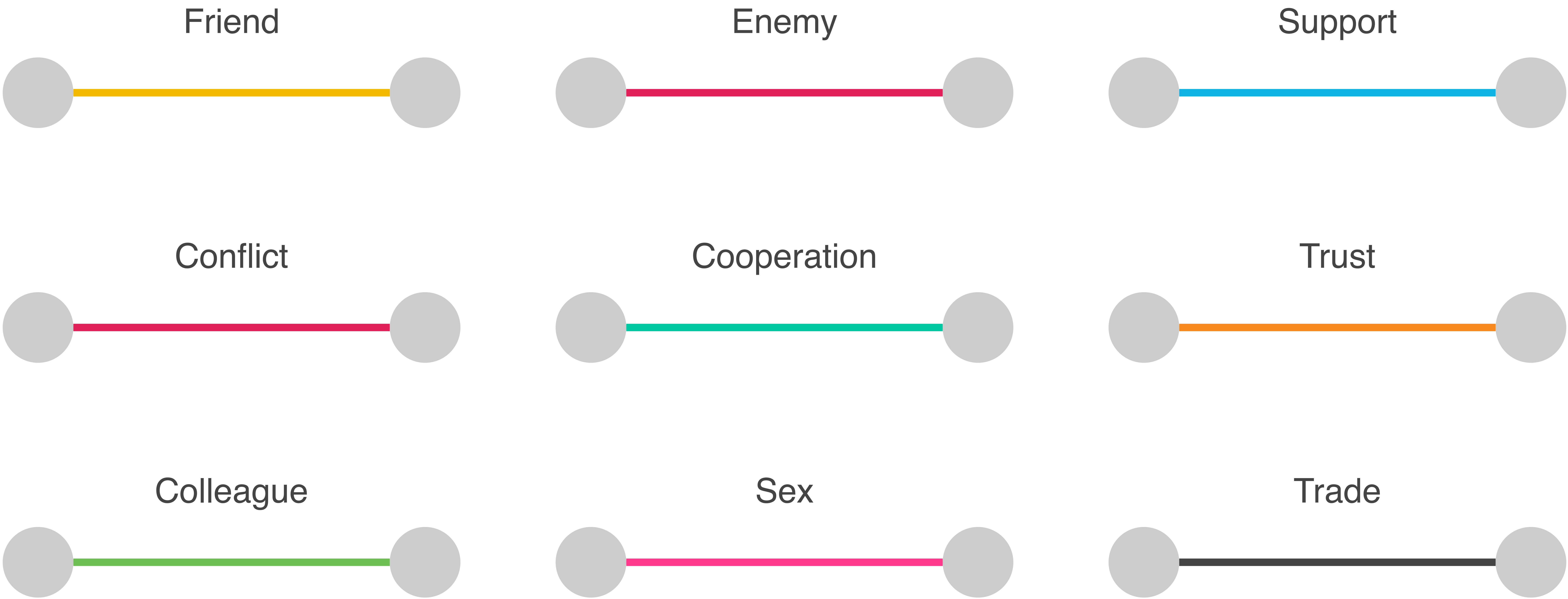
First principles

Network software with network objects



Multiple node types (user defined)

Multiple edge types (user defined)



An **interface** is an abstract 'type' of data collection instrument, designed with specific interaction affordances, which facilitates a core task within the network interview.

Example: "The categorical bin interface"

A stage is a specific instance of an interface, configured by you to collect the data needed in your interview. **Example:** "My gender categorical bin stage"

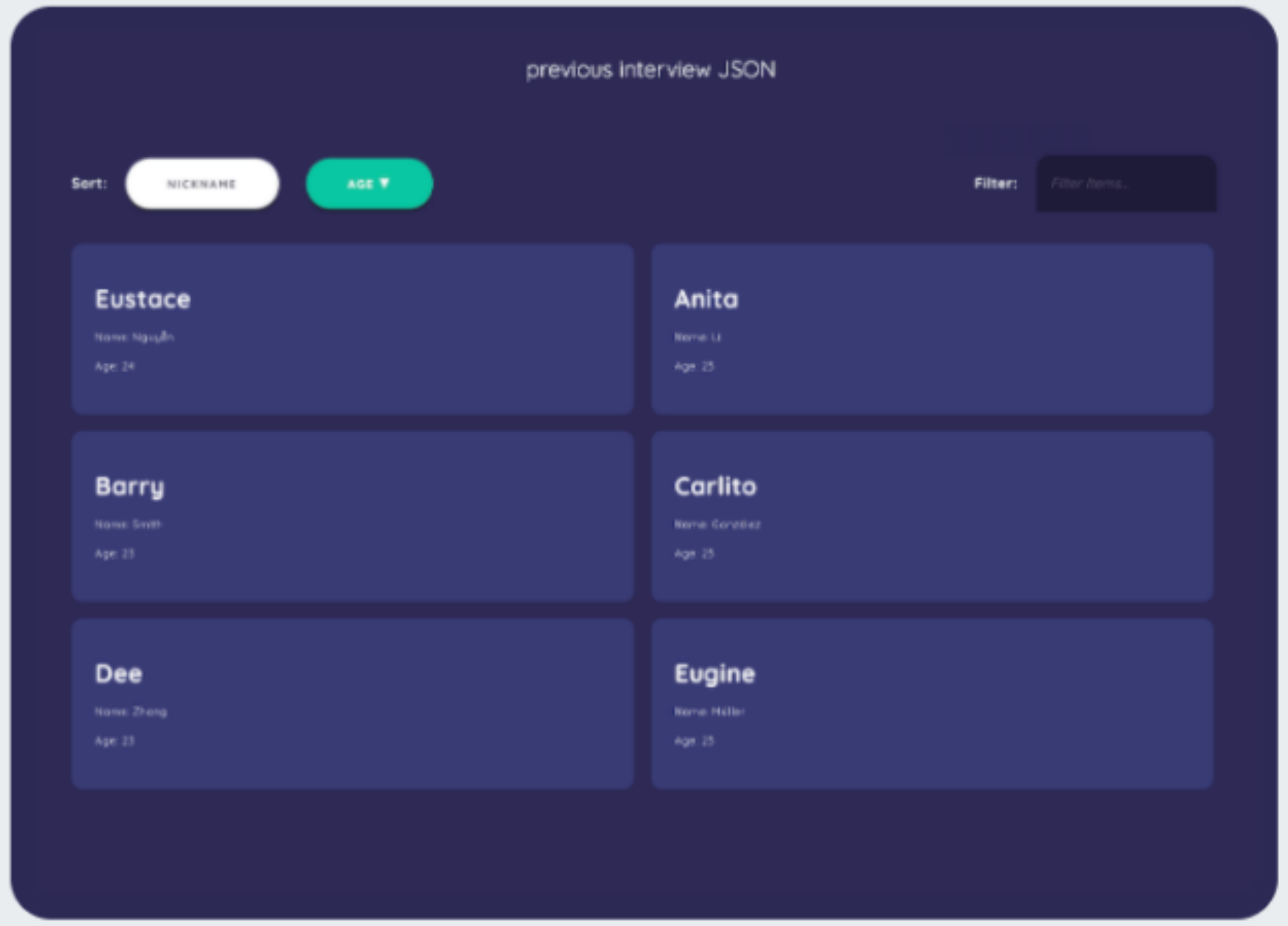
A prompt is a specific question that elicits a response. Some stages have a single prompt, some can have multiple. **Example:** "The first name generator prompt on a Name generator quick add stage"

https://documentation.networkcanvas.com/key-concepts/interfaces/

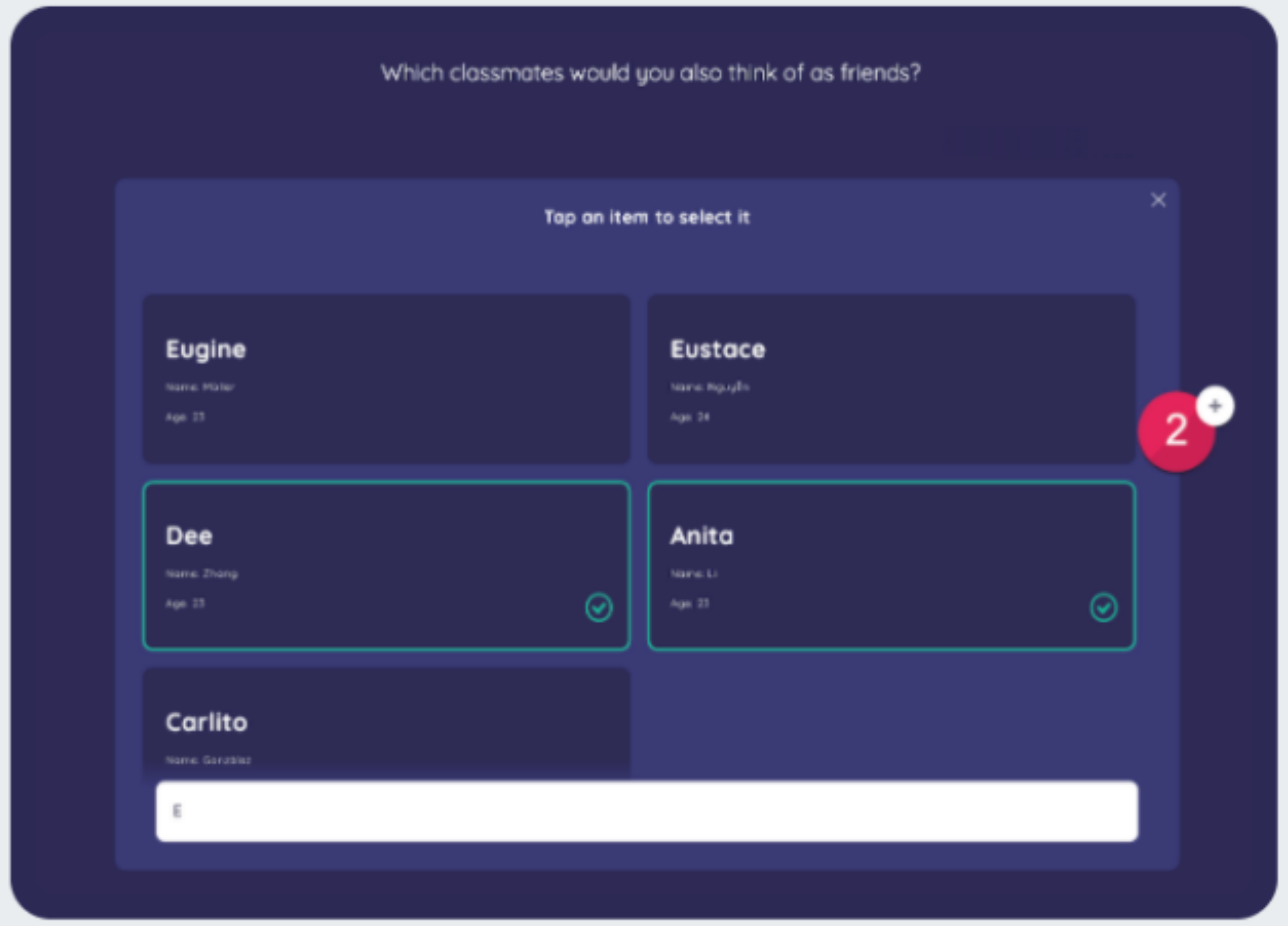
Name Generator (using forms)

Name Generator (using quick add)

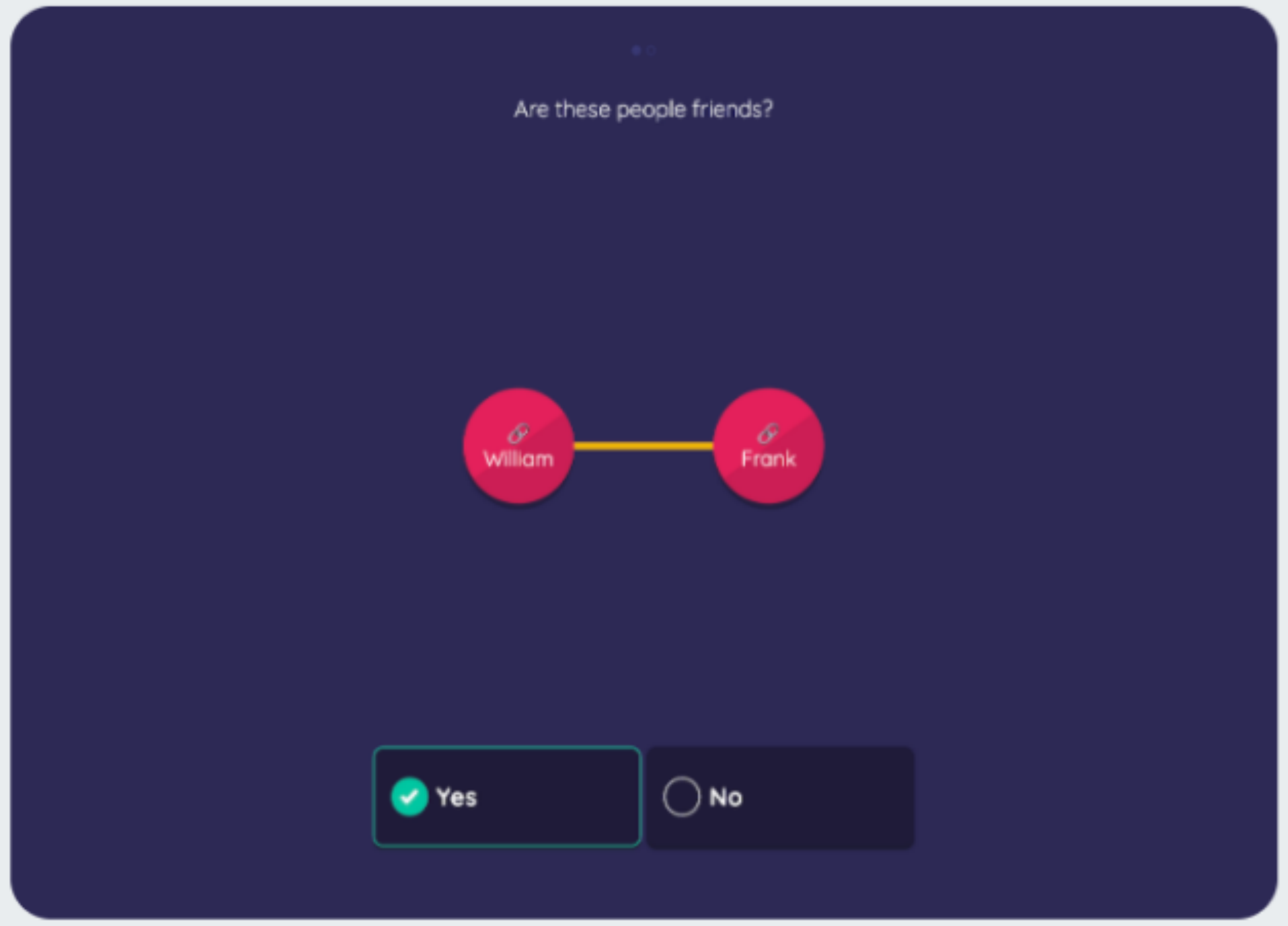
Small Roster Name Generator



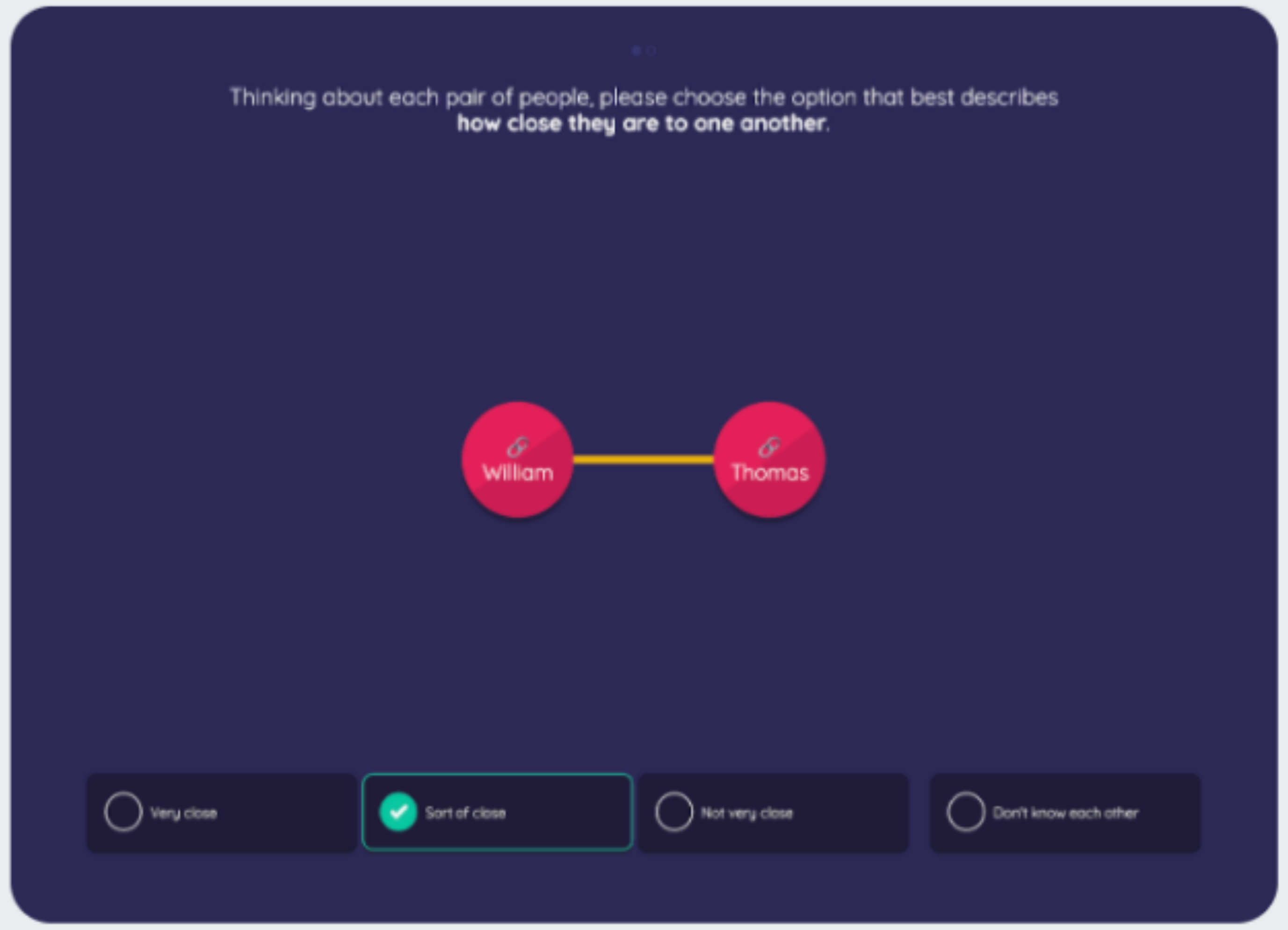
Large Roster Name Generator



Dyad Census



Tie-Strength Census



https://documentation.networkcanvas.com/key-concepts/interfaces/

Ordinal Bin

Categorical Bin

Per Alter Form

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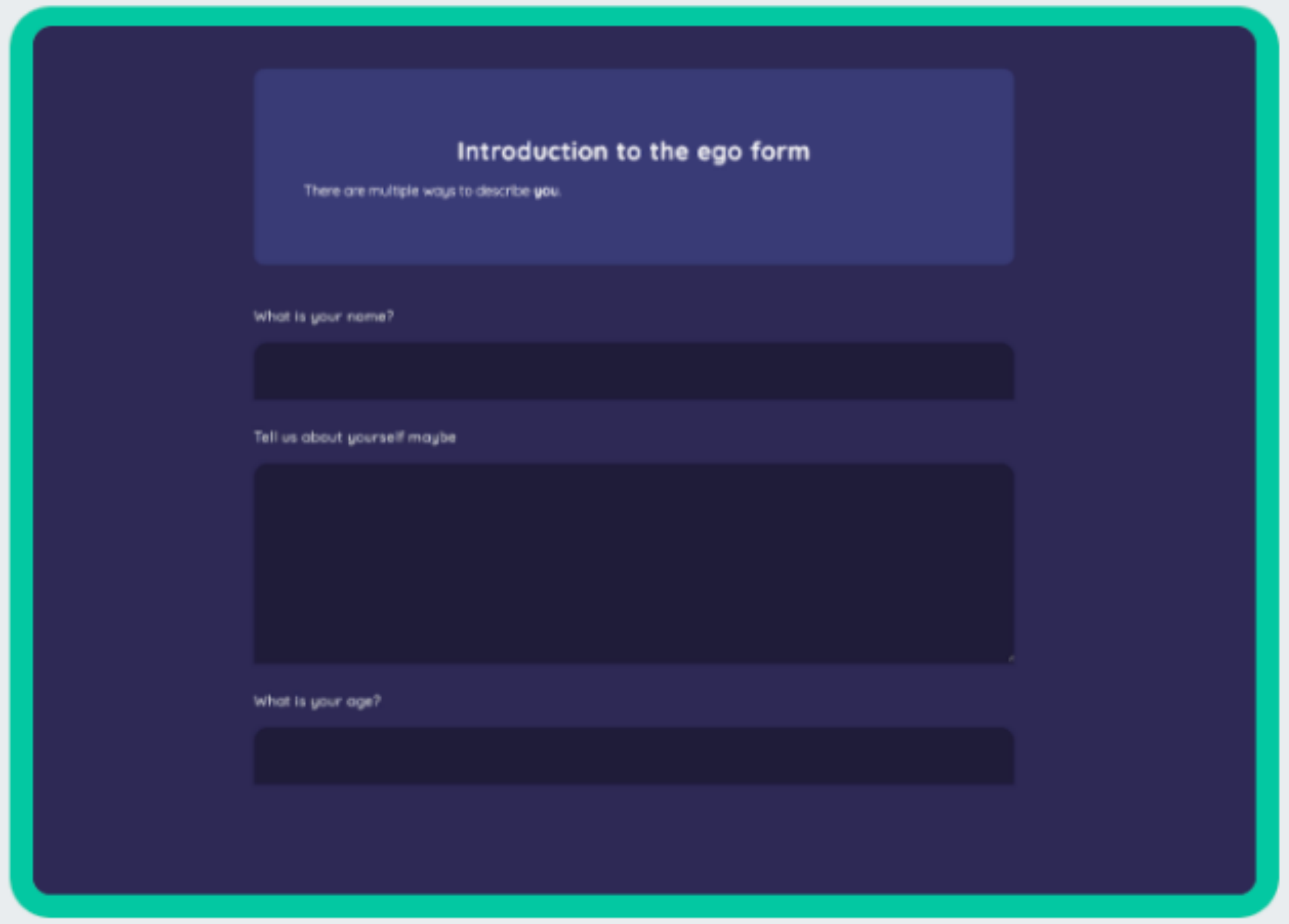
Per Alter Edge Form

Ego Form



Narrative

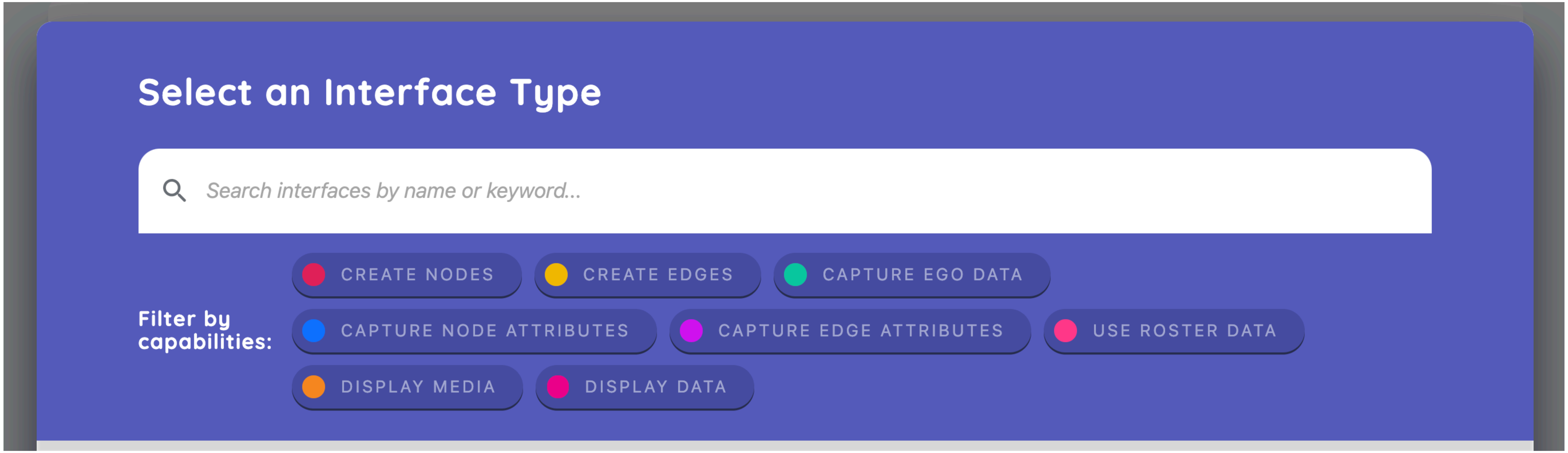
How to include network objects

Ego

Alter

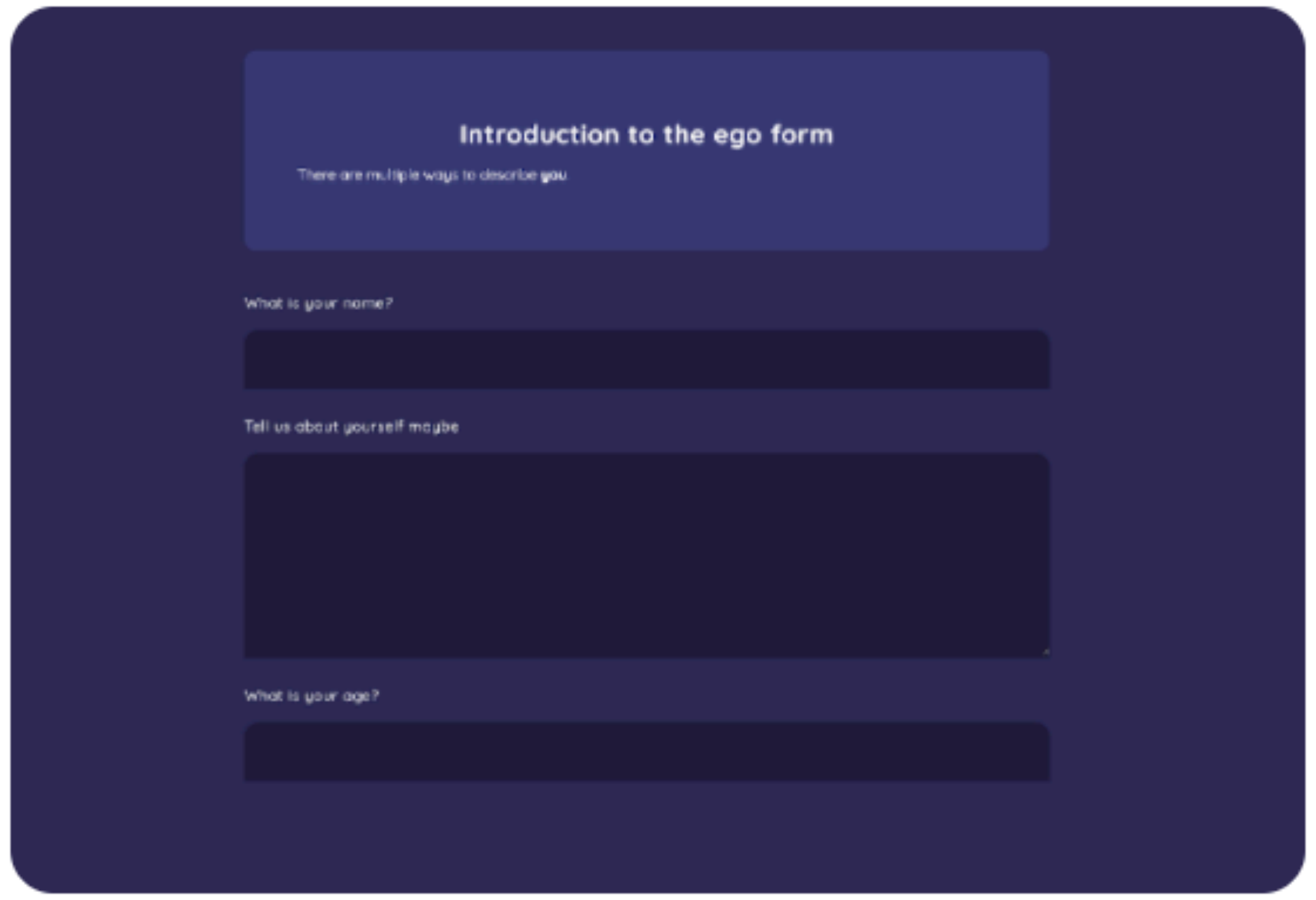
Direct tie (ego-alter)

Indirect tie (alter-alter)



Including Ego

- Generally through a participant ID (in interviewer)
- Data through an ego “form”.



Ego Form

An interface that collects data on your participant (ego).

CAPTURE EGO DATA

Including Alters

Name Generator (quick add)

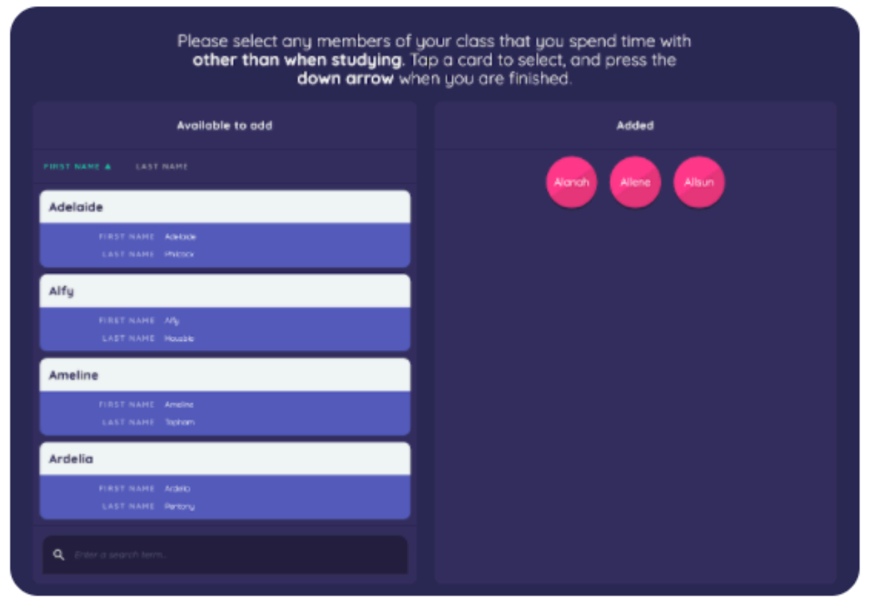
A name generator interface designed for low response-burden. Only requires a label in order to create an alter.

CREATE NODES · USE ROSTER DATA

Name Generator (using forms)

A name generator interface which provides a form that participants complete when creating an alter.

CREATE NODES · CAPTURE NODE ATTRIBUTES · USE ROSTER DATA

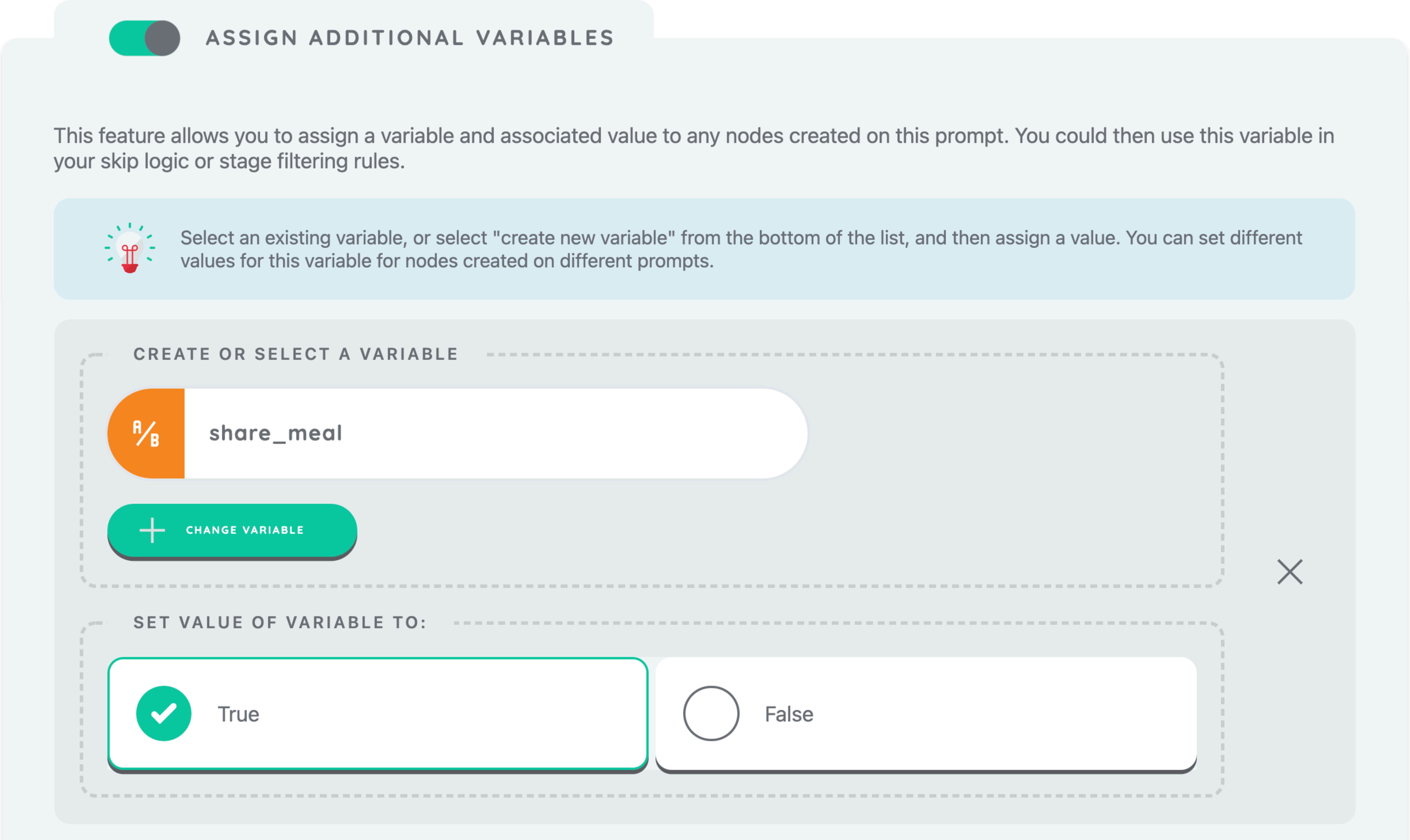


Name Generator for Roster Data

A name generator specifically for roster data, allowing sorting and filtering of the roster.

CREATE NODES · USE ROSTER DATA

Including Ego-Alter ties

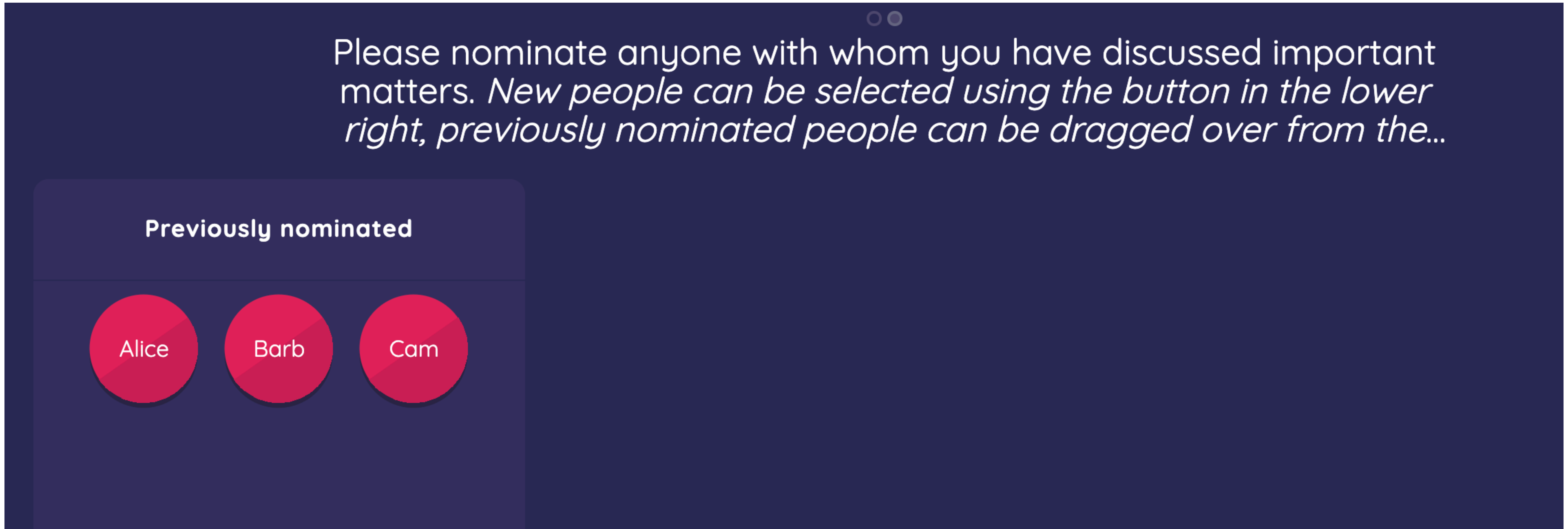


Using a side-panel

Keeping nodes consistent

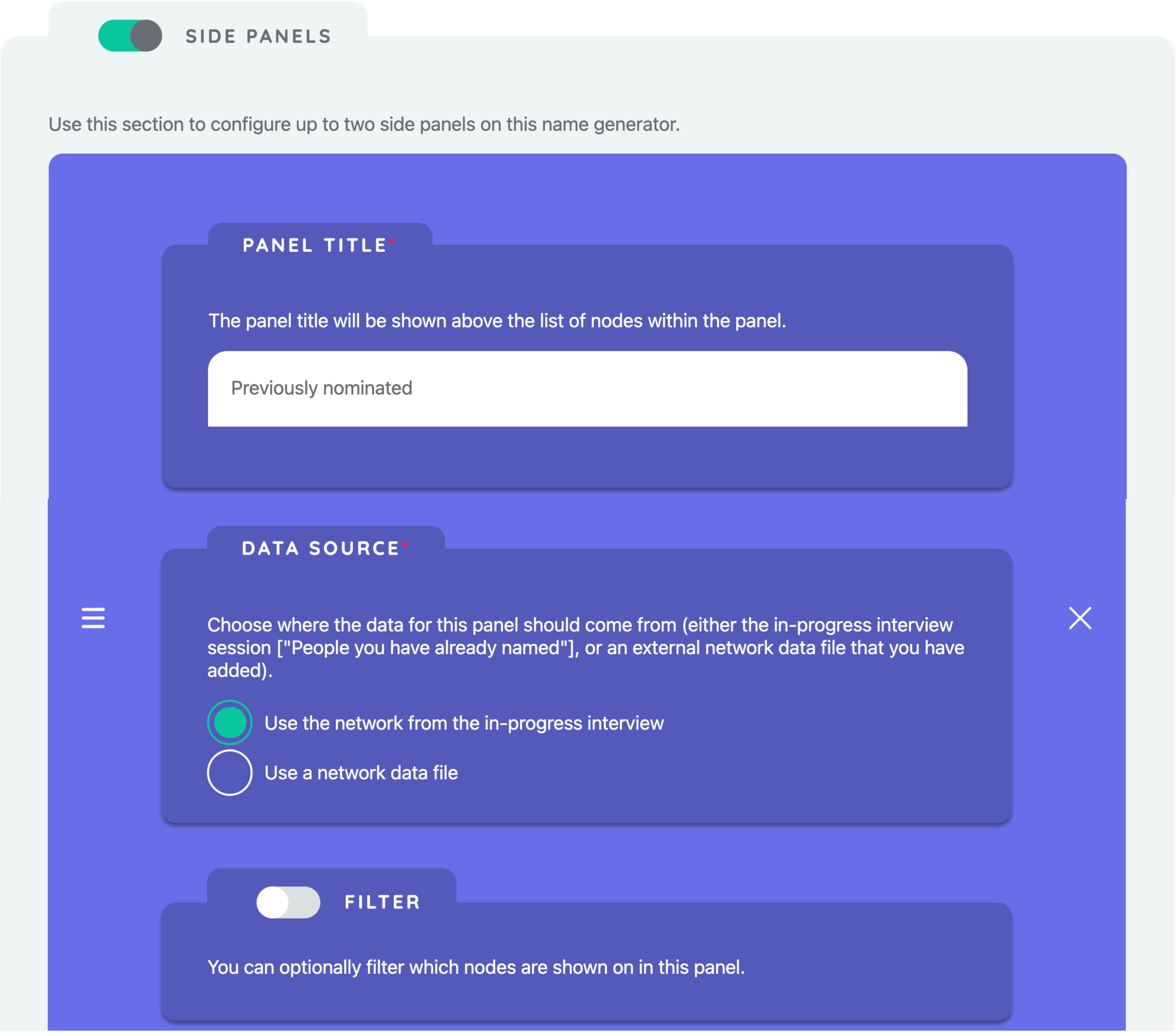
Prompt 1.

Prompt 2.



Using a side-panel

Keeping nodes consistent



Including Ego-Alter ties

PROMPTS*

Add one or more prompts below to frame the task for the user. You can reorder the prompts using the draggable handles on the left hand side.

Please nominate people that you have shared a meal with in the last month

Please nominate anyone with whom you have discussed important matters. *New people can be selected using the button in the lower right, previously nominated people can be dragged over from the panel on the left.*

CREATE NEW

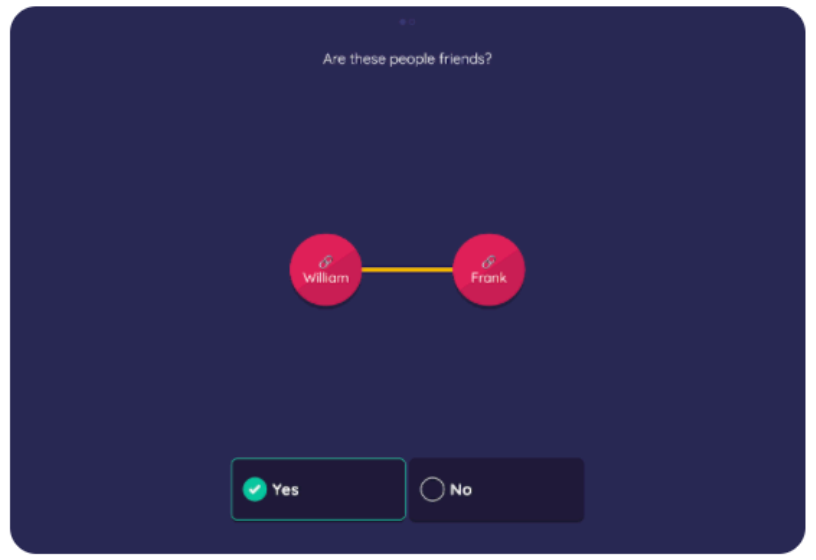
Including Alter-Alter ties

Filter by capabilities:

CREATE NODES · CREATE EDGES · CAPTURE EGO DATA

CAPTURE NODE ATTRIBUTES · CAPTURE EDGE ATTRIBUTES · USE ROSTER DATA

DISPLAY MEDIA · DISPLAY DATA



Dyad Census

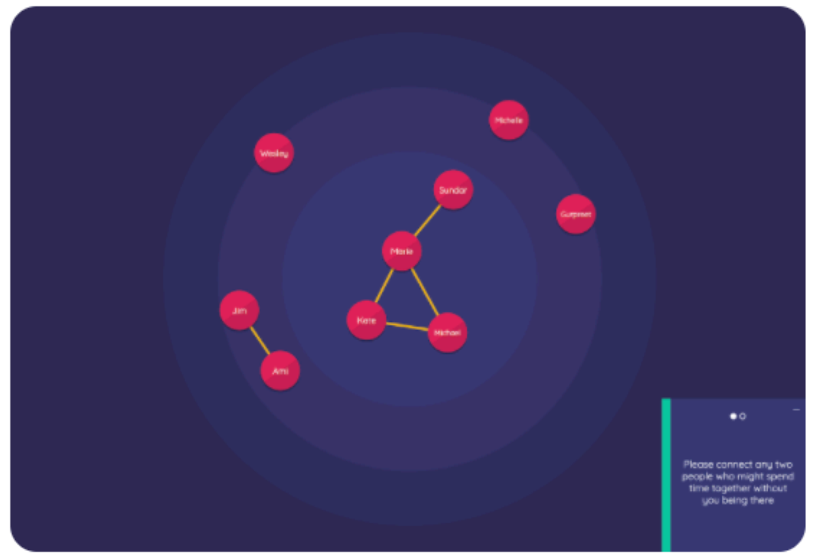
A name interpreter interface that creates edges by systematically surveying all alters in the interview network.

CREATE EDGES

Tie-Strength Census

Combines a dyad census with an ordinal variable to simultaneously capture the strength of ties between alters.

CREATE EDGES · CAPTURE EDGE ATTRIBUTES



Sociogram

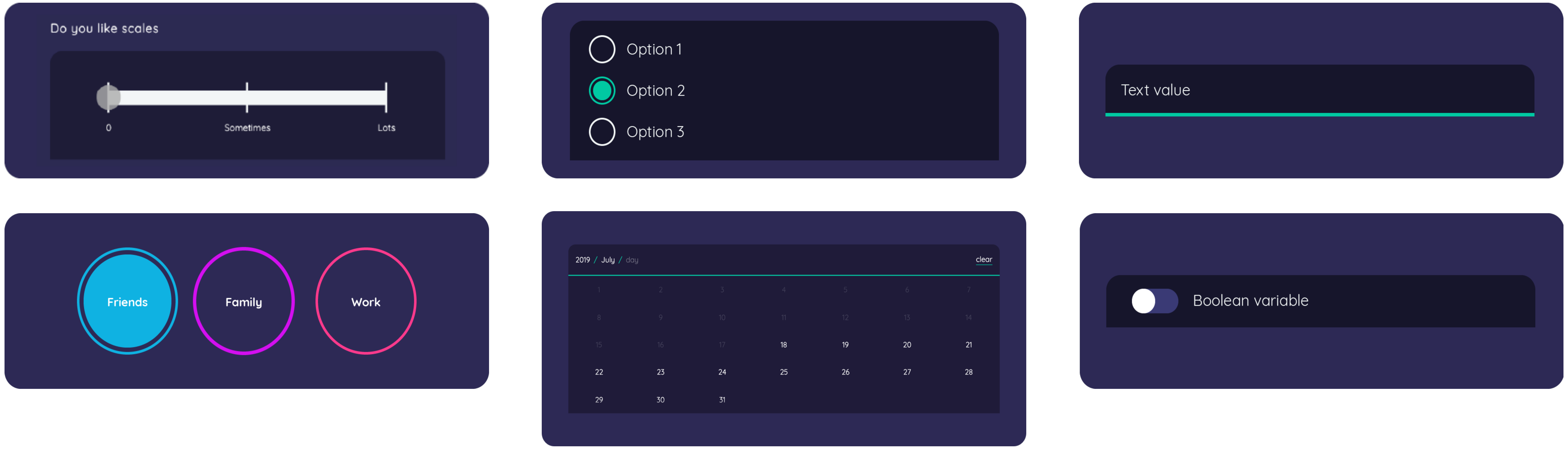
Designed for spatially arranging alters (either manually or automatically), creating edges between them, and highlighting the presence of alter attributes.

CREATE EDGES · CAPTURE NODE ATTRIBUTES · DISPLAY MEDIA

Including data about network objects

Text	Basic short text string,
Number	Basic whole number (integer),
Ordinal	Ordered categorical data, where each value is described by a label and a value. Only supports single values,
Categorical	Unordered categorical data where each value is described by a label and a value. Supports multiple values.
Boolean	True or False values,
Layout	X/Y coordinate values, normalized between 0 and 1 in each axis.
DateTime	Date variable capable of storing resolution down to the day.
Scalar	A continuous floating point number between 0 and 1

Data type comes from an input control

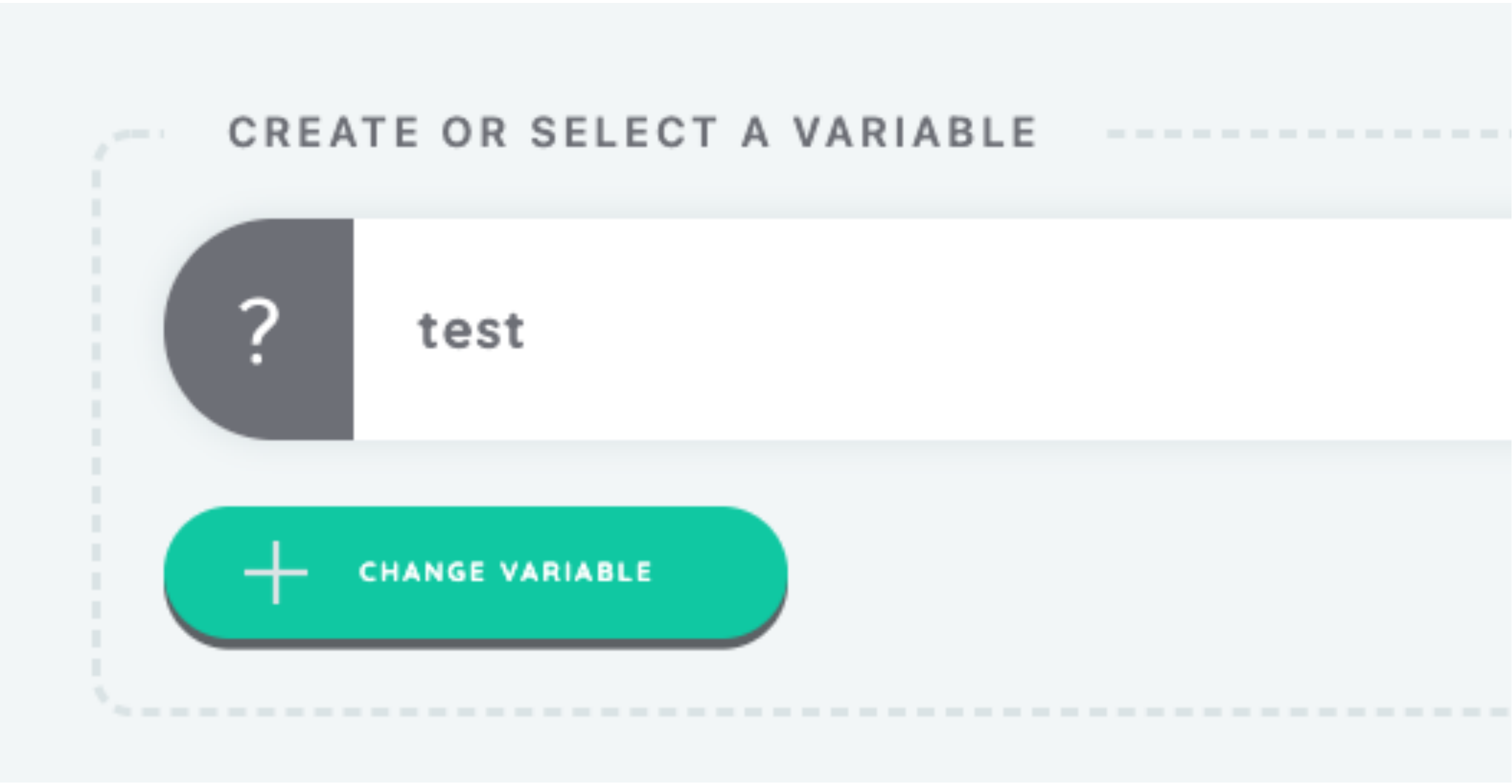


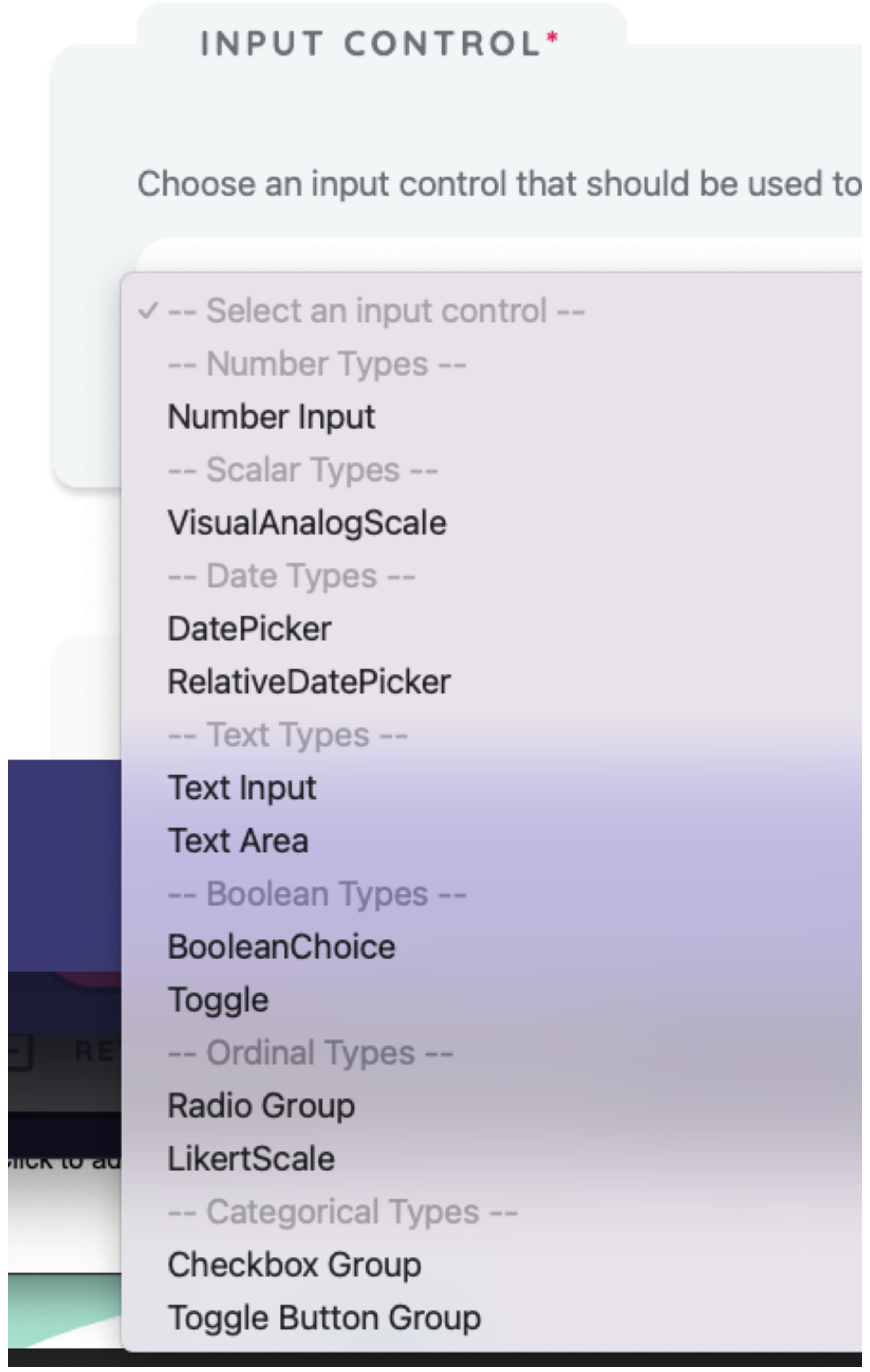
Different input controls will have different features, like the ability for a calendar to start from 'today' or have a specific date range.

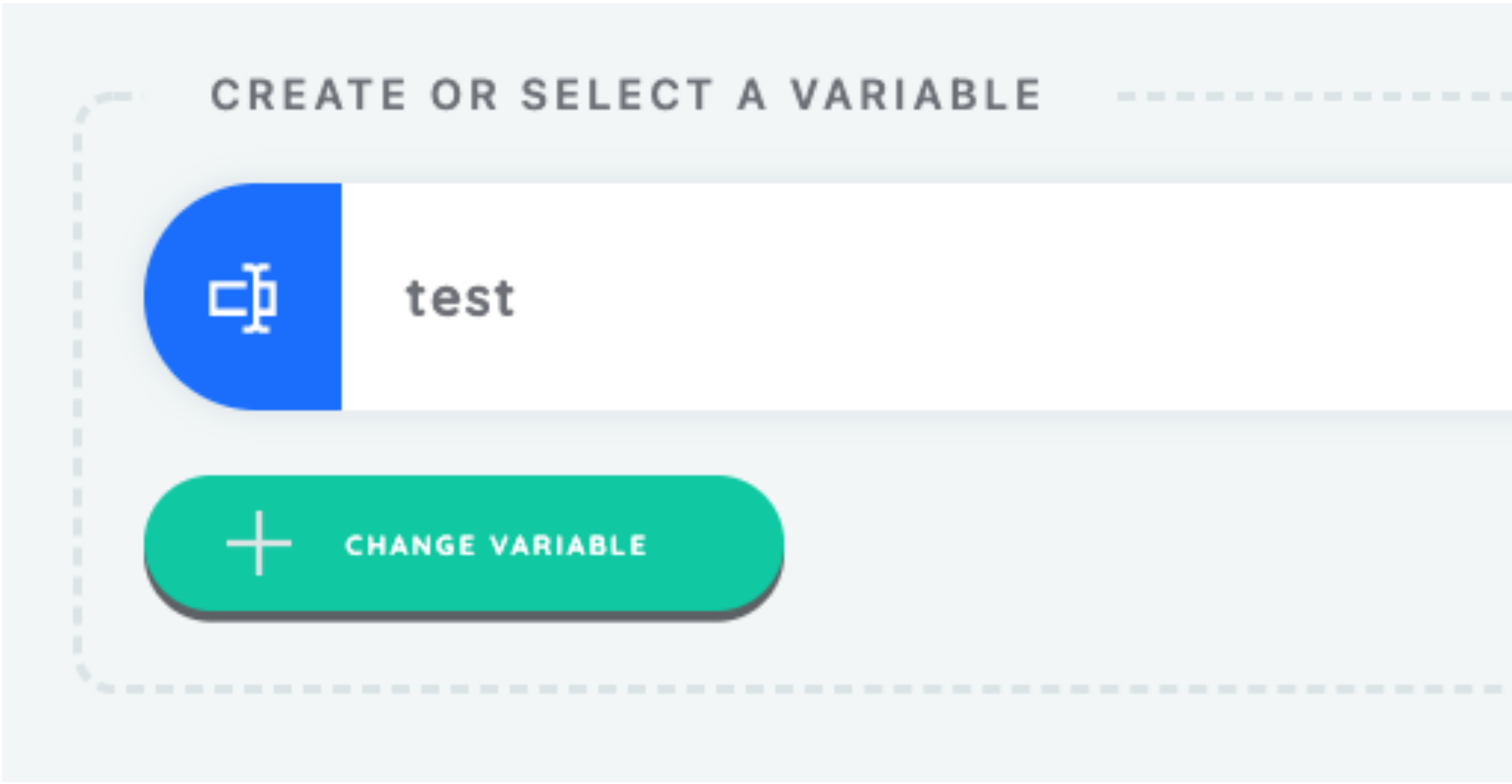
The input controls are all listed under key concepts in the documentation.

https://documentation.networkcanvas.com/key-concepts/input-controls/

Variables get a name, then an input type







Ordinal Bin

Categorical Bin

Per Alter Form

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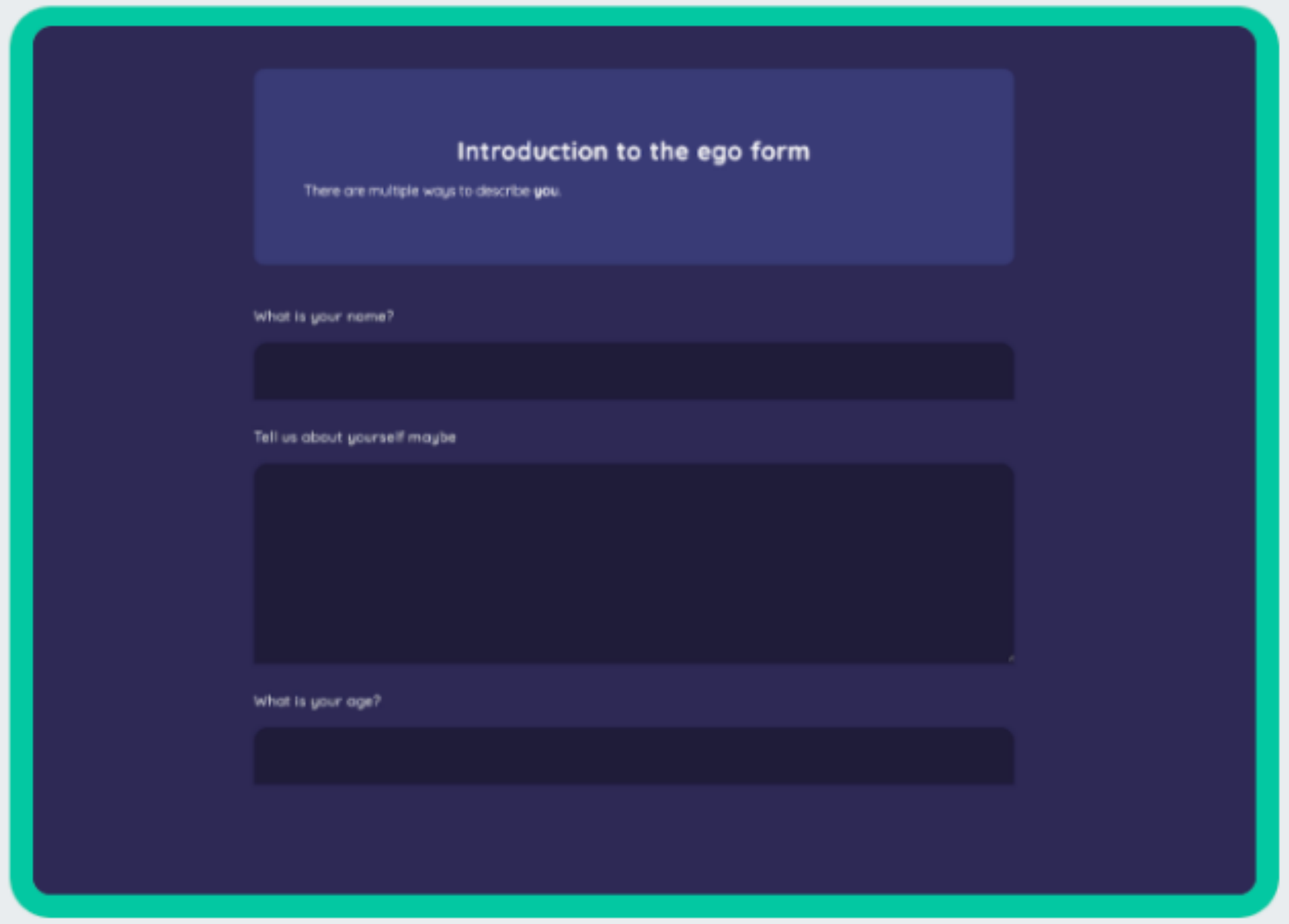
Per Alter Edge Form

Ego Form



Narrative

Assigning data in steps

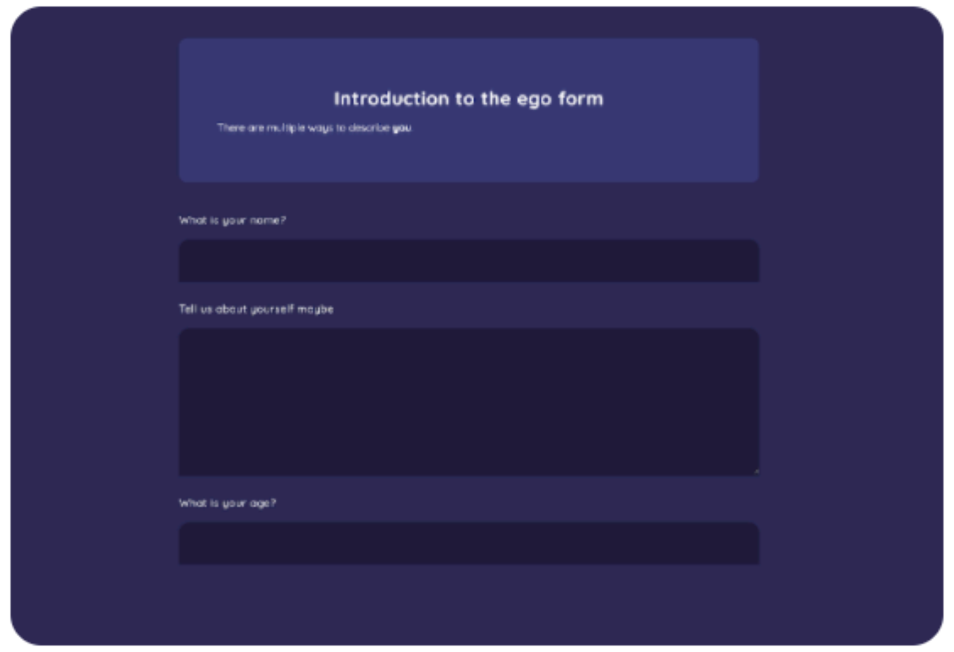
A general approach

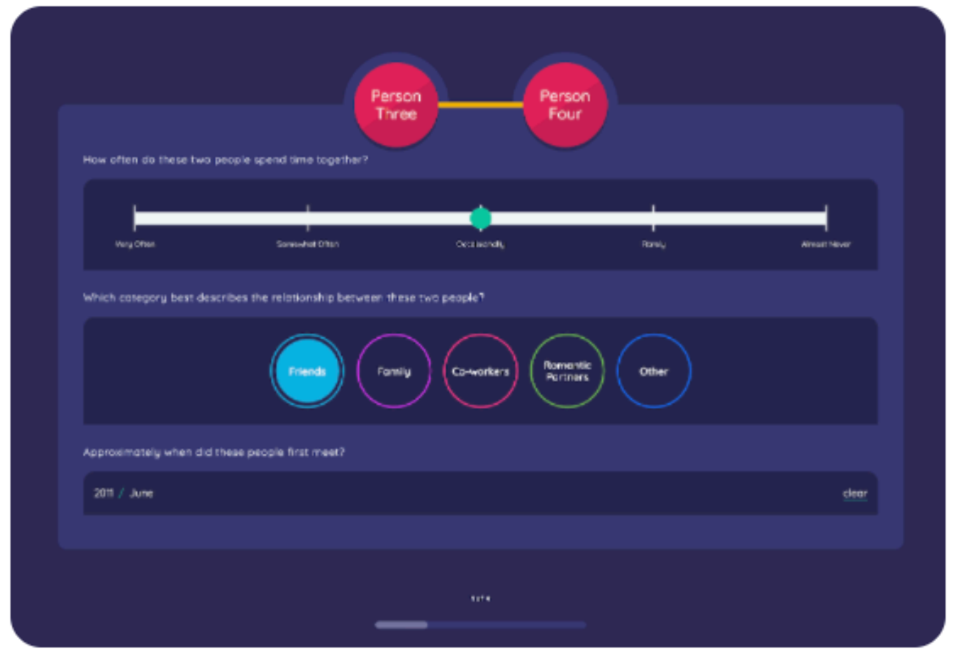
- Select a Stage based on whether you are collecting Ego, Node, or Edge data;
- Include the type of ego, node, or edge;
- Create a variable for that data;
- Select an input control;
- Network Canvas will then assign the variable a type based on that control.

Selecting the data collection "approach"

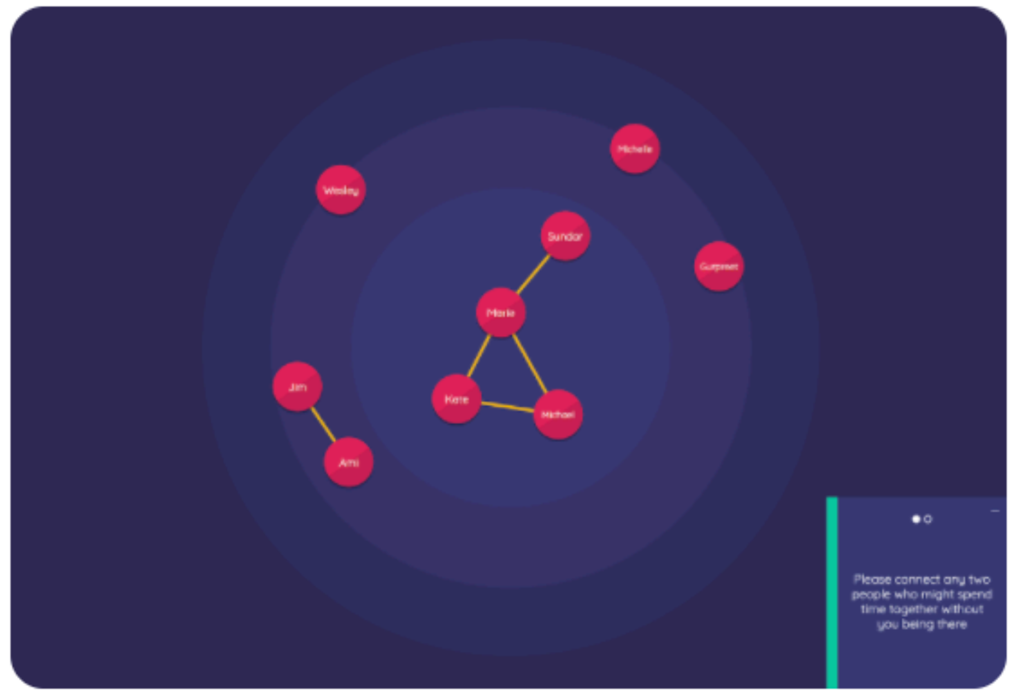
Per network approaches versus per object approaches

Per object:
- Ego
- Alter
- Edge





Per network:
- Bins
- Sociograms



Considering the following questions

Maybe I’m just hungry

“How often do you go for a meal with this person”,

“Select some cuisines that you would eat with this person”

“To the best of your knowledge, is this person a vegetarian or vegan?”

- What are the data types that we are expecting?
- What input controls would work for which data types?
- Should these be per network or per alter?

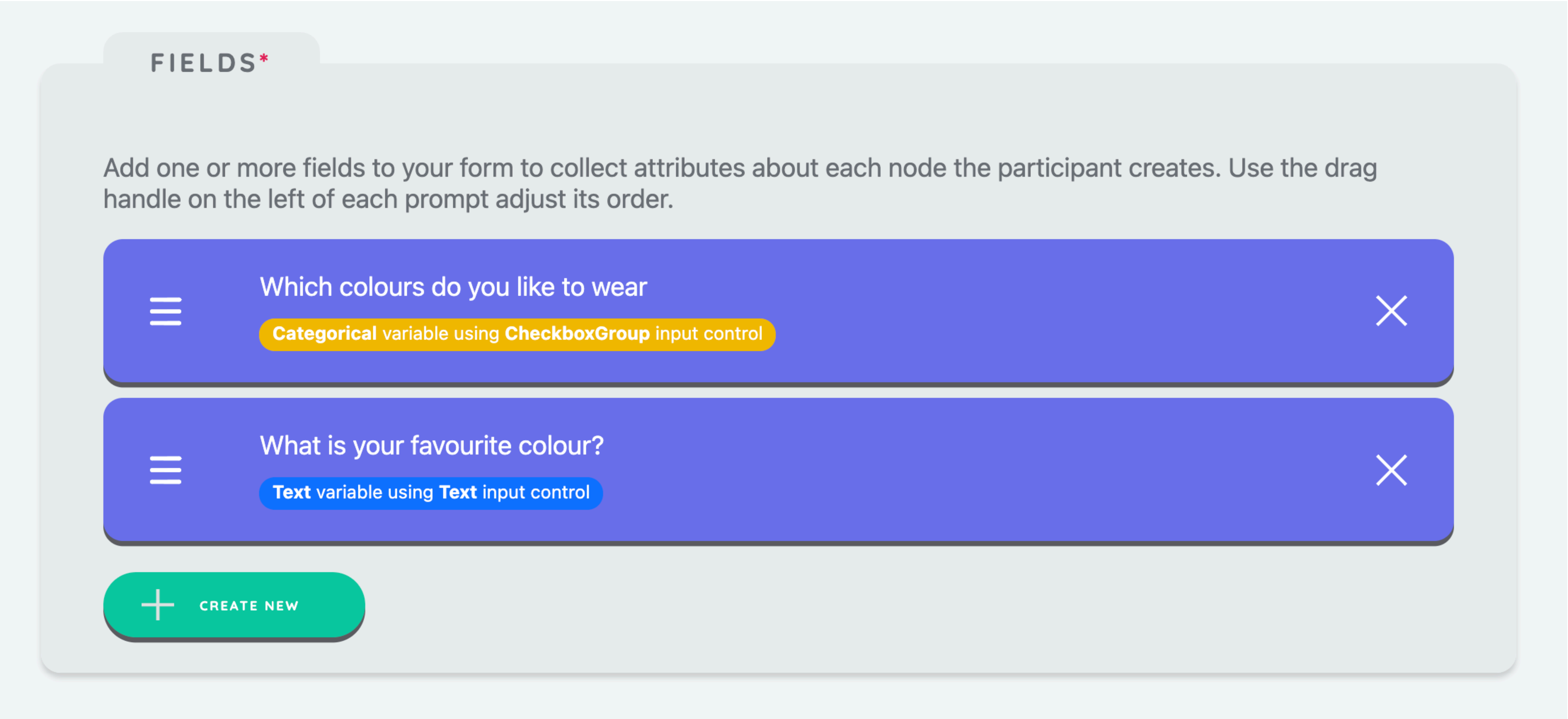
Network Canvas Gotchas

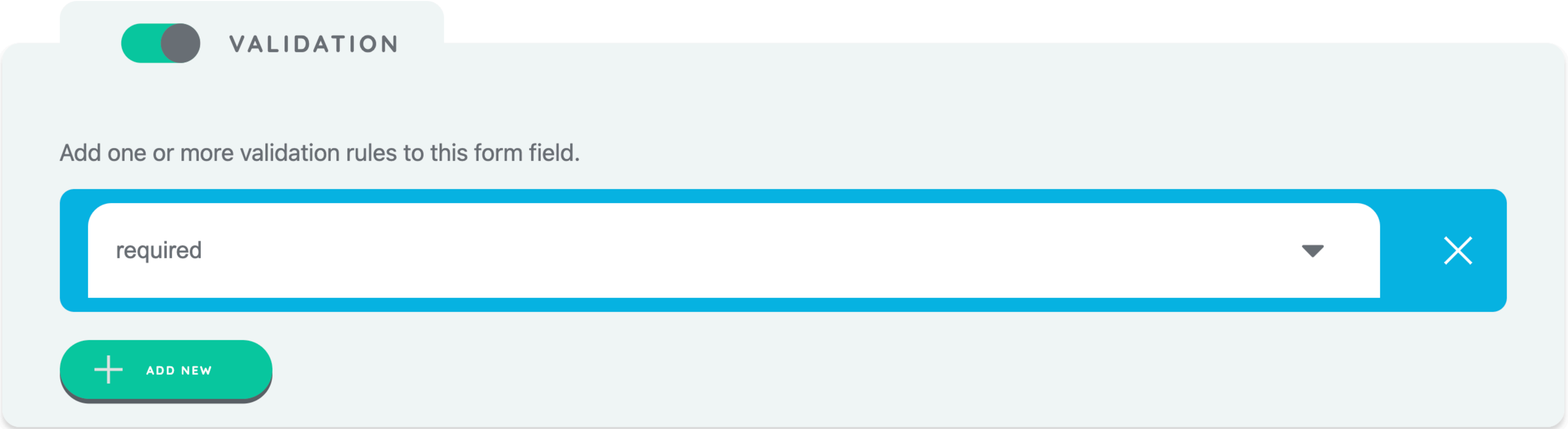
Some things about style learned from experience

- Form length and missing data
- Duplicate nodes
- Boring interviews

Dealing with form length

- Example from SB22_architect_demo.netcanvas
- Warnings and mandatory questions
- You can also use validation options to help this.





Duplicate nodes

- Side panels which can be used for these things to remind people of previously mentioned alters.
- It’s important to think of node types as constitute separate alter pools that you draw upon in different stages and prompts.
- Make sure people do not nominate a node twice or three times.
- You can also set minimum and maximum number of alters.

Boring interviews

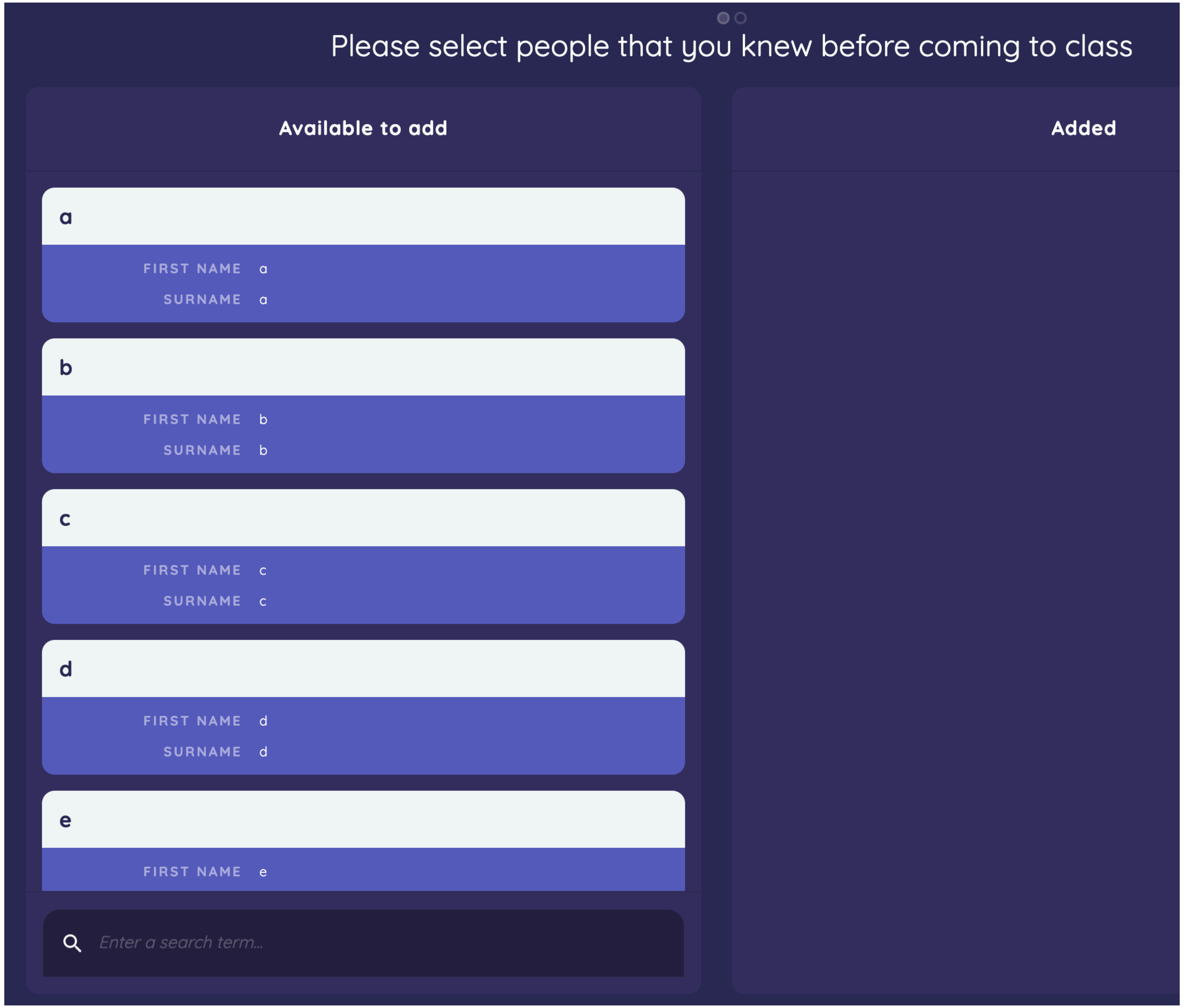
Try to keep it fresh

- Use the fact that variables are passed around in order to break up the interview.
- Remember that a layout variable is not stage-specific so you can alternate with sociograms and other questions.
- Think about the ways in which you are asking the participant. Asking about disparate questions of an alter or a cohesive questions? Think per network forms or per alters.
- Break up forms, don't just use validation.

Ideas for whole network work

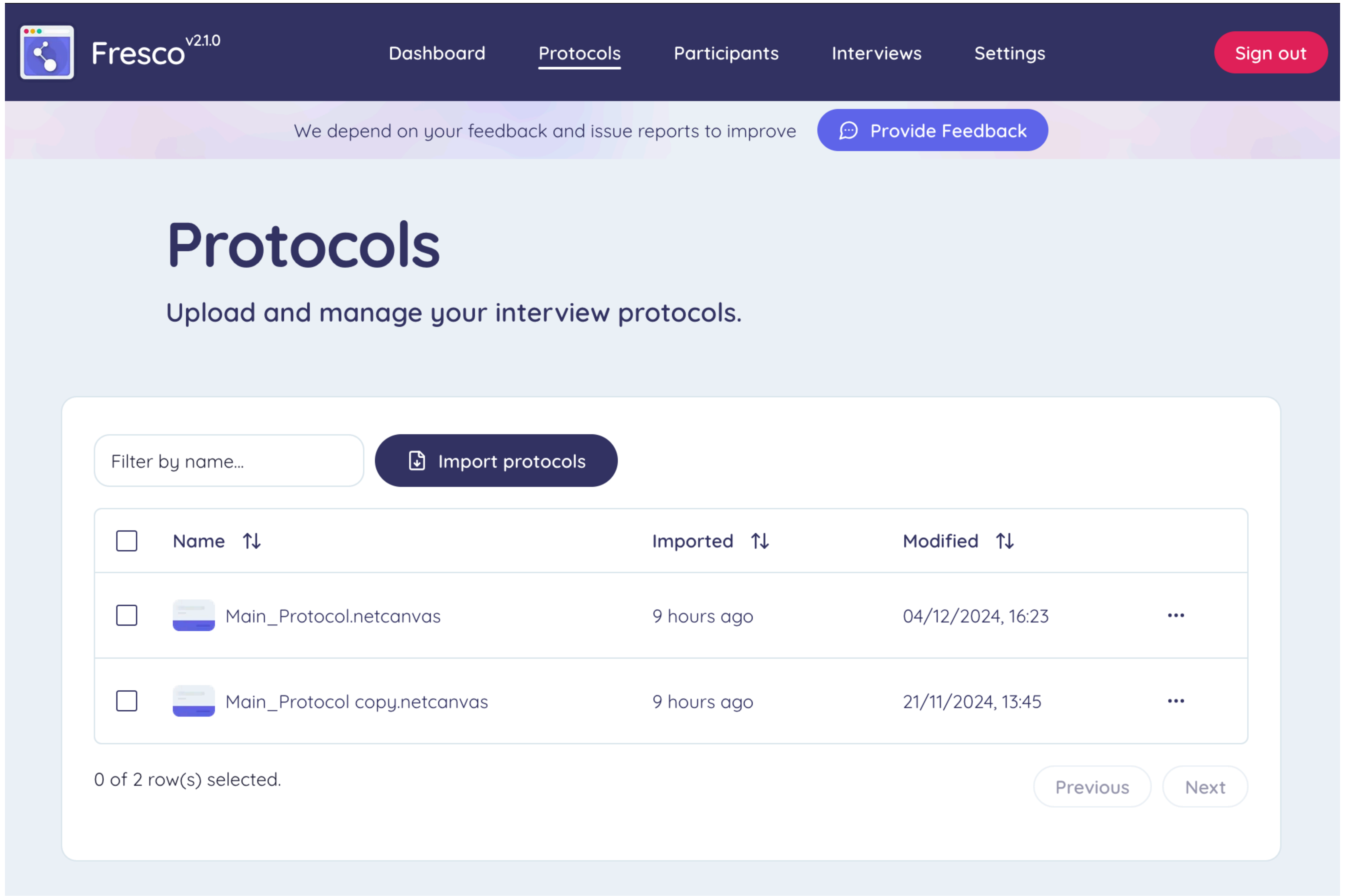
Using Rosters

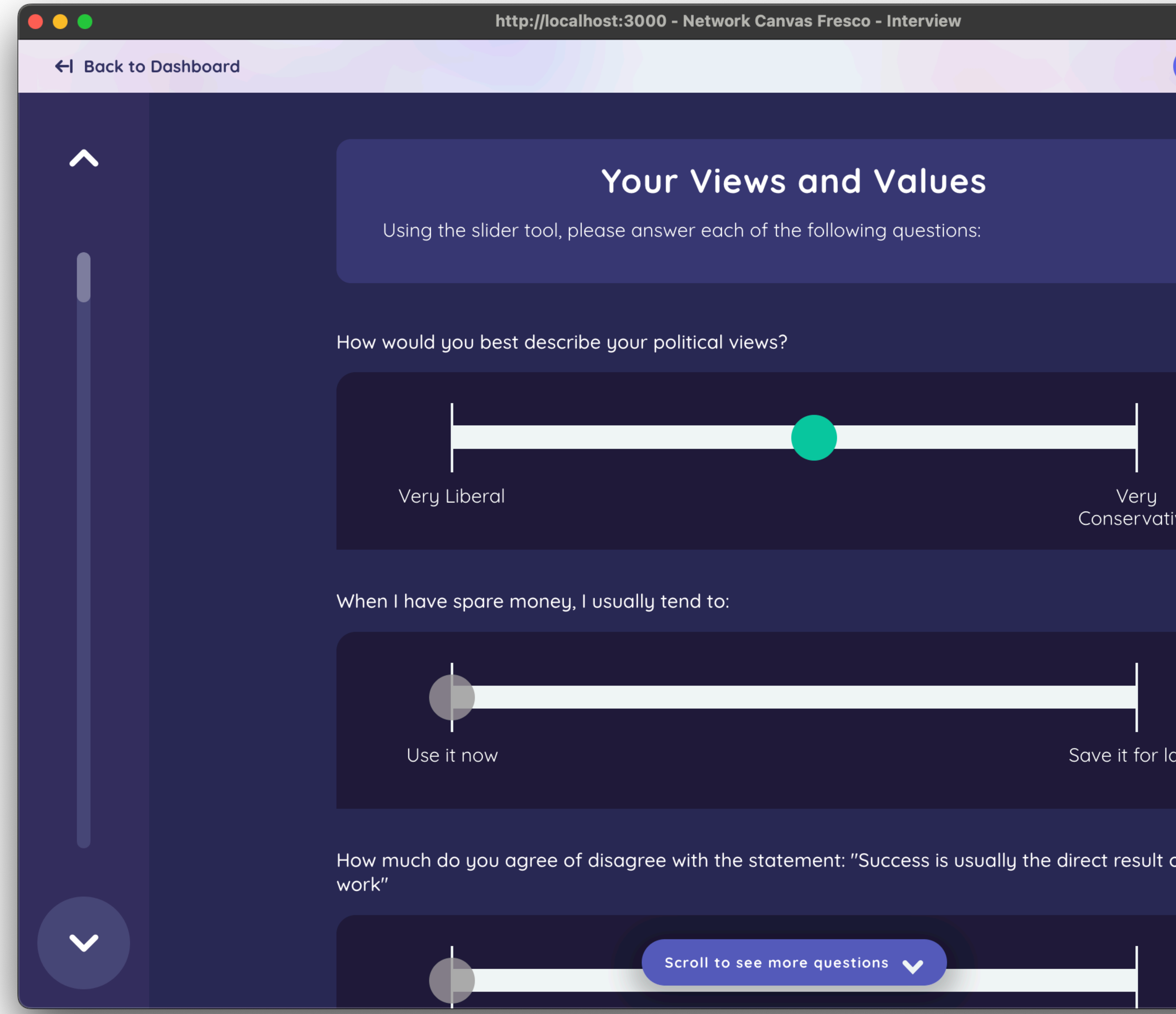
- See the demonstration using `SB22_whole_network_demo.netcanvas`
- You can upload a roster as a CSV file. Be sure to have a column with name, preferably in the first column.
- Let's open in Architect to see the previous screen.



New this year and upcoming

Introducing Fresco!





Notes about Fresco

It is _not_ as easy to install or configure

- Fresco is a web portal that is built with similar components as Interviewer but needs a server to run on.
- The documentation provides an example way to do this using a third party server, and the readme provides a way to do local hosting using software called Docker.
    - This involves a little familiarity with GitHub (and ideally some JavaScript).
- We understand this. Our medium term goal (i.e. within two years) is a platform “Network Canvas Studio” where you can host network canvas surveys.

Now - Into the Protocols!

- We have three demonstration protocols to consider.
- I will walk through the first one with everyone to also show how to install a protocol.
    - Importantly, I will be talking through these in ARCHITECT while also doing these in INTERVIEWER.

Your turn!

Thank you so much

- What sort of tasks did you want to accomplish with Network Canvas?
- What challenges have you had?

Network Canvas

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